

The context-dependent epigenetic and organogenesis programs determine 3D vs. 2D cellular fitness of MYC-driven murine liver cancer cells

Reviewed Preprint

v2 • April 17, 2025

Revised by authors

Reviewed Preprint

v1 • January 6, 2025

Jun Yang , Jie Fang, Shivendra Singh, Brennan Wells, Qiong Wu, Hongjian Jin, Laura Janke, Shibiao Wan, Jacob Steele, Jon Connelly, Andrew Murphy, Ruoning Wang, Andrew Davidoff, Margaret Ashcroft, Shondra Pruett-Miller

St. Jude Children's Research Hospital Memphis United States • University of Nebraska Medical Center Omaha United States • Abigail Wexner Research Institute at Nationwide Children's Hospital Columbus United States • University of Cambridge Cambridge United Kingdom

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This manuscript provides potentially **important** findings examining in 2D and 3D models in MYC liver cancer cells changes in DNA repair genes and programs in response to hypoxia. The authors use **convincing** methodology in most cases, but there is some concern that the analysis is **incomplete**.

<https://doi.org/10.7554/eLife.101299.2.sa2>

Abstract

3D cellular-specific epigenetic and transcriptomic reprogramming is critical to organogenesis and tumorigenesis. Here we dissect the distinct cell fitness in 2D (normoxia vs. chronic hypoxia) vs 3D (normoxia) culture conditions for a MYC-driven murine liver cancer model. We identify over 600 shared essential genes and additional context-specific fitness genes and pathways. Knockout of the VHL-HIF1 pathway results in incompatible fitness defects under normoxia vs. 1% oxygen or 3D culture conditions. Moreover, deletion of each of the mitochondrial respiratory electron transport chain complex has distinct fitness outcomes. Notably, multicellular organogenesis signaling pathways including TGF β -SMAD, which is upregulated in 3D culture, specifically constrict the uncontrolled cell proliferation in 3D while inactivation of epigenetic modifiers (*Bcor*, *Kmt2d*, *Mett13* and *Mettl14*) has opposite outcomes in 2D vs. 3D. We further identify a 3D-dependent synthetic lethality with partial loss of *Prmt5* due to a reduction of *Mtap* expression resulting from 3D-specific epigenetic reprogramming. Our study highlights unique epigenetic, metabolic and organogenesis signaling dependencies under different cellular settings.

Introduction

Cellular fitness is determined by a constellation of genetic, epigenetic, metabolic and environmental factors that dynamically interact and communicate¹. Identifying the genes and pathways controlling cellular fitness not only enhances understanding of the pathophysiological mechanisms, but also helps to define potential therapeutic targets to treat human disorders. Genome-wide CRISPR and shRNA screenings are powerful genetic approaches that are invaluable for identifying fitness genes^{2–6}. While some studies have applied genome-wide or focused CRISPR library screenings using 3D organoids^{7–11}, which better recapitulate *in vivo* tissue features, most genome-wide screenings have been performed using 2D cultured cells grown in normoxia (normoxia or normoxic conditions tested refer to the standard ambient air oxygen concentrations used in conventional *in vitro* models). However, direct comparison of cell fitness in 3D vs 2D culture is currently lacking. Two recent studies have identified several hundred oxygen concentration-dependent fitness genes^{12,13}, demonstrating that environmental factors such as oxygen have a profound impact on cellular fitness. We previously showed that organoids in normoxic culture induced a strong hypoxia signature in cells¹⁴, indicating that 3D culture may mimic 2D hypoxic culture, at least to some degree. Oxygen is an essential molecule that fostered the evolution of monocellular to multicellular organisms. Oxygen is primarily utilized by mitochondria as the final electron acceptor to generate ATPs in the oxidative phosphorylation pathway, as well as a variety of chemical reactions in cells^{15–17}. Physiological oxygen concentrations can vary anywhere from 0.5% in the large intestine to just under 21% in the trachea, with most major organs operating around 3–7% oxygen^{13,15,17}. To maintain cellular fitness, cells have evolved oxygen sensing mechanisms to adapt oxygen variation by altering expression of a plethora of genes, mainly through hypoxia-inducible factors (HIFs) and epigenetic modifiers such as histone demethylases some of which are HIF targets^{17–19}. HIF is a heterodimer composed of HIF-1a or HIF-2a (also named EPAS1) and their partner HIF-1b (also known as ARNT)¹⁸. While HIF-1b is constitutively expressed, HIF-a availability is tightly regulated by prolyl hydroxylases (PHD) and the von Hippel Lindau (VHL) tumor suppressor protein¹⁸. At ambient oxygen levels, HIF-a is hydroxylated by oxygen-dependent PHDs leading to its rapid proteasomal degradation by the VHL E3 ubiquitin ligase complex. Conversely, hypoxic conditions render HIF-a available for dimerization with HIF-1b. The HIF-a/HIF-1b complex then translocate to the nucleus, binds to the hypoxia response elements (HRE), and drives gene expression. The dysregulation of HIF-PHD-VHL pathway is associated with many human disorders including cardiovascular diseases and cancers^{20,21}.

Organogenesis is an orchestrated process that is regulated by developmental signaling pathways mediated by TGFb, Wnt, Hedgehog, Notch, and others. Genetic alterations in these signaling pathways cause cancers and other human diseases²². Solid tumors are recognized as functionally abnormal organs²³, and they are classically under a heterogeneous state of hypoxia in part due to insufficient and pathologic angiogenesis²⁴. The dynamic cell-cell and cell-environment interactions of tumor cells in 3D suggest an added layer of complexity to our current understanding of tumor biology²³. Clonal monoculture studies do not fully capture the hypoxia- and 3-dimensional-driven mechanisms found in heterogeneous 3D tumors²³. Single-cell RNA-seq analysis of 1163 human tumor samples covering 24 tumor types shows that hypoxia, together with *MYC* gene clusters are the two of the most commonly recurring transcriptional programs in heterogeneous tumors²⁵. Identifying fitness genes of *MYC*-driven cancers in different oxygen levels and in 3D conditions may reveal context-dependent vulnerabilities that can be exploited for therapies.

Here, we show that under 3D culture conditions, cells undergo unique epigenetic and transcriptomic reprogramming of gene transcription, particularly for the genes regulated by TGFb-SMAD signaling. Further, we use genome-wide CRISPR/Cas9 gene-editing technology to

knock out every gene in a MYC-driven cancer cell line cultured as a monolayer under normoxia and 1% oxygen, or in 3D spheroids under normoxia, followed by comprehensive comparative analyses of cellular fitness genes under these three conditions. We show that knockout of *Vhl* leads to a fitness defect in normoxic 2D conditions but not in 1% oxygen monolayer and 3D spheroids. Conversely, deletion of *Hif1a* or *Hif1b* causes fitness defects in 1% oxygen and 3D spheroids but not in a normoxic 2D conditions. Cells tolerate loss of mitochondrial ribosomal genes better in 1% oxygen than normoxia. However, knockout of mitochondrial respiratory complex I-V causes distinct cellular fitness outcomes for each complex under these three conditions. Knockout of genes in organogenesis signaling pathways such as TGFb-SMAD specifically leads to a growth advantage of 3D spheroids while inactivation of epigenetic modifiers (*Bcor*, *Kmt2d*, *Mettl3* and *Mettl4*) in monolayer and 3D spheroids results in opposite fitness outcomes, and these genes often function as tumor suppressors in human cancer. We also discover distinct metabolic requirements for fatty acid and cholesterol synthesis in the three conditions. We further identify a context-dependent synthetic lethality of 3D culture conditions with partial loss of *Prmt5* due to a reduction of *Mtap* expression, resulting from 3D-specific epigenetic reprogramming of the *Mtap* gene. Our study reveals distinct epigenetic and metabolic dependencies of cancer cells in different environments, highlighting the critical role of organogenesis signaling pathways in regulating tumor cell growth.

Results

Transcriptomic reprogramming of cancer cells under 1% oxygen in 2D or normoxic 3D culture conditions

We have recently generated a MYC-driven liver cancer genetic model (ABC-Myc mouse) and derived cell lines such as NEJF10²⁶, which are readily cultured in 2D and 3D conditions (Fig. 1-figure supplement 1A), and which are suitable for genome-wide genetic screenings²⁶. In comparison with human cancer cell lines which usually have multiple genetic defects, the ABC-Myc cell lines have a simpler genotype (MYC alone as a driver), which may be less impacted by epistasis when used for genome-wide identification of fitness genes. To understand the gene transcription differences in normoxia vs hypoxia, and 2D vs 3D culture, we performed RNA-seq after NEJF10 cells were cultured in normoxia and 1% oxygen in monolayer, or normoxic 3D spheroids for 48 hours. We performed pairwise comparisons of genes upregulated and downregulated in monolayers grown under hypoxic (1%) or normoxic (21%) conditions to those of a culture grown as 3D spheroids under normoxia (21%). We found that 44.8% (925 out of 2065) and 49.4% (1021 out of 2065) of genes that were upregulated and downregulated respectively under 1% oxygen were shared with the culture grown as 3D spheroids (Fig. 1-figure supplement 1B, Supplementary file 1 and 2). The most significant gene signatures upregulated in 3D vs 2D under 21% oxygen are hypoxia-related (Fig. 1-figure supplement 1B), consistent with our previous observation that organoid culture induced a strong hypoxia signature¹⁴. REACTOME and KEGG pathway enrichment analysis revealed that hypoxia and 3D culture induced common genes involved in focal adhesion, proteoglycans, extracellular matrix and receptor interaction, as well as oncogenic signaling pathways such as Hippo, HIF1, Wnt and PI3K-AKT²⁷, but inhibited the expression of genes involved in mitochondrial translation and the proteasome (Fig. 1C, left; Fig. 1-figure supplement 1C). However, genes involved in protein translation or ribosome biology, oxidative phosphorylation and protein export were dominantly downregulated by hypoxia (Fig. 1C, right; Fig. 1-figure supplement 1C), while 3D culture enhanced the expression of genes in TGF- β -SMAD signaling, regulation of pyruvate dehydrogenase (PDH) complex, NOTCH, MAPK, autophagy, and lysosome pathways (Fig. 1C, middle; Fig. 1-figure supplement 1C). 3D culture also suppressed the expression of genes in the DNA homologous recombination and Fanconi anemia pathway (Fig. 1C, middle; Fig. 1-figure supplement 1C), which are involved in DNA repair and frequently mutated in various cancers²⁸.

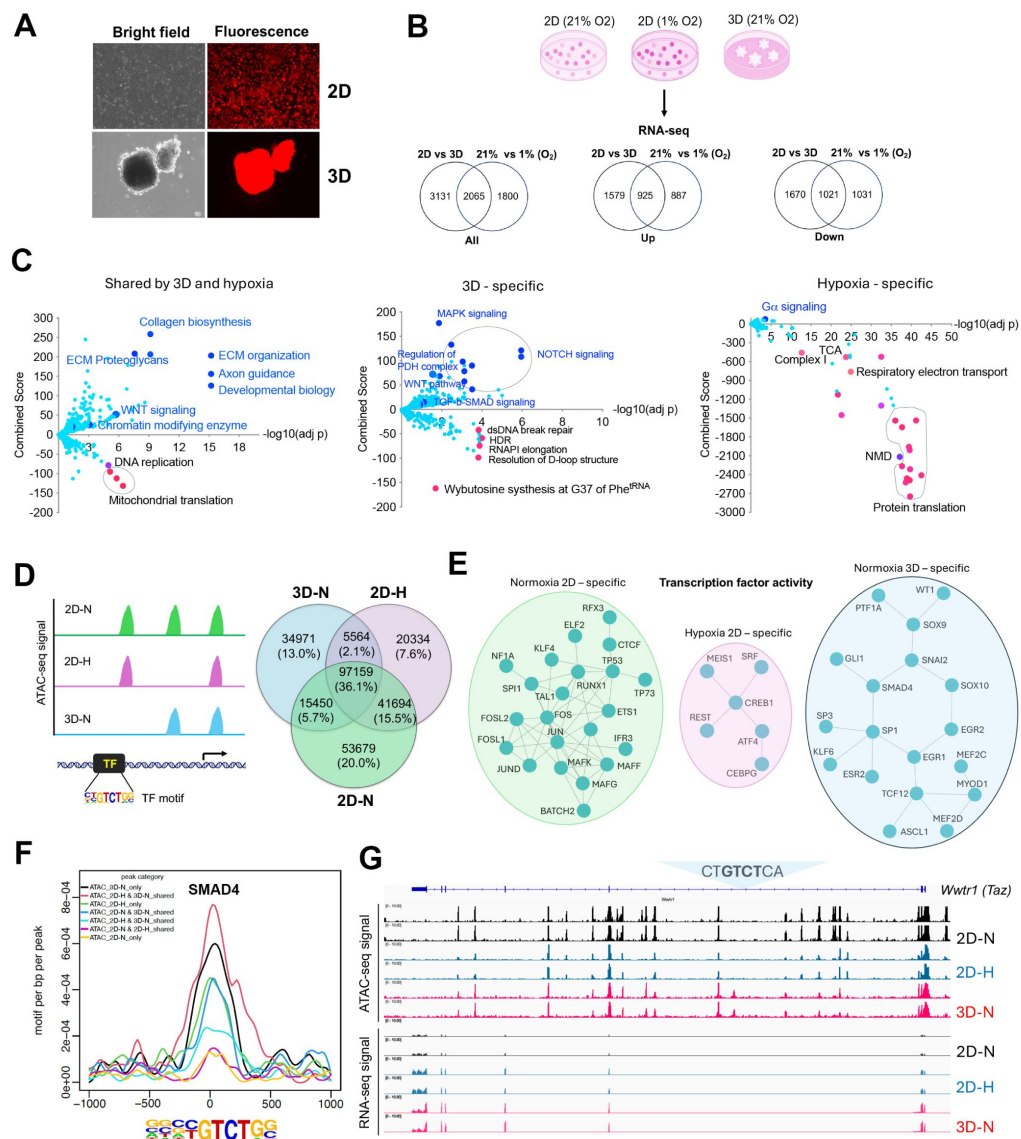


Figure 1.

Transcriptomic and epigenetic reprogramming in hypoxic 2D and normoxic 3D conditions.

(A) The ABC-Myc driven hepatoblastoma organoids serve as a good model for genome wide fitness screening. The ABC-Myc lineage was validated through expression of TdTomato.

(B) RNA-sequencing data from cells cultured in our conditions of interest are represented in Venn Diagrams of genes in 3D vs 2D normoxia and 2D-Hypoxia vs 2D-Normoxia ($n = 3$ per group). The leftmost Venn diagram represents a total count of all genes identified, followed by the Up and down labeled diagrams reporting the upregulated and downregulated genes, respectively. The numbers in the Venn diagrams represent the number of significant genes expressed in the respective conditions.

(C) The volcano plots, generated using Enrichr, show significant REACTOME pathways shared by cells in normoxic 3D and hypoxic 2D (left), specific to hypoxic 2D (middle) and normoxic 3D conditions (right) ($n = 3$ per group). The X-axis represents the significance of expression change for a gene in $-\log_{10}$ (adjusted P-value). The Y-axis represents the

combined score for enrichment of a given pathway with positive values indicating enrichment and negative values indicating a decrease in the pathway.

(D) ATAC-seq analysis for cells under normoxic 2D, hypoxic 2D and normoxic 3D conditions (n=2 per group). The left cartoon indicates the changes of ATAC-seq signals under different culture conditions and predicted transcription factor binding motif. Right Venn diagram showing the common and unique ATAC-seq signals (numbers and percentages) under three conditions.

(E) Protein-protein interaction network (STRING confidence threshold = 0.7) formed by transcription factors that are predicted to be highly active under normoxic 2D, hypoxic 2D and normoxic 3D, respectively.

(F) SMAD4 motif densities around ATAC-seq open chromatin regions (± 1000 bp) by categories. Motif density was determined by HOMER (Hypergeometric Optimization of Motif EnRichment) program and normalized to that in a background sequence of equal length. X-axis indicates the distance to peak center (bp).

(G) IGV snapshot shows the ATAC-seq signals and RNA-seq reads under normoxic 2D, hypoxic 2D and normoxic 3D conditions. The “CTGTCTCA” SMAD4 binding motif lies within the ATAC-seq peak specifically appears in normoxic 3D conditions.

Alternative splicing is an important mechanism cells use to adapt to cellular stresses by producing different gene isoforms²⁹. We therefore compared the five major splicing events (alternative 3' splicing, alternative 5' splicing, mutually exclusive exon usage, intron retention and exon skipping) induced by hypoxia and 3D culture (Supplementary file 3–5). In comparison with normoxia in 2D culture, hypoxia dominantly induced intron retention, while 3D cultures dominantly induced exon skipping (**Fig. 1-figure supplement 1D**). Exon skipping was also the major event in 3D culture when compared with 2D hypoxic cultures (**Fig. 1-figure supplement 1D**). These data indicate that splicing is an important mechanism for cellular adaptation to environmental stress, as evidenced by the altered splicing events of pyruvate dehydrogenase kinase 1 (*Pdk1*) gene under different culture conditions (**Fig. 1-figure supplement 1E**). Exon skipping in exons 7 and 8 of *Pdk1* specifically occurred in 3D culture, while an alternative 5' splicing event only occurred in hypoxic conditions (**Fig. 1-figure supplement 1E**). Pathway analysis of the five splicing events under three conditions demonstrated that various biological processes (i.e., ribosome, spliceosome, metabolic pathways, and DNA repair) are involved in all conditions (**Fig. 1-figure supplement 1F**, Supplementary file 6–8), indicating an overly complex regulation of post-transcriptional processing. Taken together, these data demonstrate that cells undergo commonly shared and contextdependent transcriptional reprogramming in normoxia, hypoxia, and 3D spheroid conditions. Particularly, pathways involved in multicellular organogenesis, tissue homeostasis and a variety of human diseases are specifically induced by hypoxia and/or 3D culture, suggesting that conventional 2D culture under normoxia may not recapitulate the cellular fitness in 3D culture or under hypoxic conditions.

Context-specific epigenetic reprogramming of cells in hypoxic 2D and normoxic 3D conditions

To determine the mechanism by which culture conditions induced context-specific transcriptomes, we performed an assay for transposase-accessible chromatin with sequencing (ATAC-Seq) for determining chromatin accessibility across the genome in normoxic and hypoxic 2D and normoxic 3D culture conditions. The results showed a drastic effect of culture conditions on DNA accessibility (**Fig. 1D**). While 36.1% of chromatin accessibility regions were shared in three conditions, unique chromatin accessibility was observed (20% for 2D normoxia, 7.6% for 2D hypoxia and 13% for 3D). Nevertheless, the chromatin accessibility peaks showed similar distributions at transcription start sites, intragenic regions, upstream and downstream of gene regions (**Fig. 1-figure supplement 2A**). Then we performed Homer motif analysis to predict the

activity of transcription factors in these conditions. Pairwise comparisons (normoxic 2D vs. hypoxic 2D, normoxic 2D vs. normoxic 3D, hypoxic 2D vs. normoxic 3D) of transcription factors followed by protein-protein interaction network analysis demonstrated that unique transcription factor modules functioned under these conditions (**Fig. 1E**, **Fig. 1-figure supplement 2B-D**). While AP1 (FOS, FOSL1, FOSL2, JUN, JUND) and CTCF were more active in normoxic 2D culture, ATF4 was more selectively active in hypoxic 2D conditions. ATF4 is known to be involved in regulating the integrated stress response and cell survival under hypoxic conditions³⁰. However, under normoxic 3D culture, the activity of transcription factors such as SMAD4, SOX9, GLI1 and WT1 was specifically high (**Fig. 1E**, **Fig. 1-figure supplement 2B-D**). The high transcriptional activity of SMAD4 under 3D (**Fig. 1F**) was consistent with the upregulation of target genes of TGF- β -SMAD signaling in normoxic 3D culture (**Fig. 1C, middle**). *Wwtr1*, encoding a transcriptional cofactor TAZ downstream of the Hippo pathway, is a well-characterized target of TGF- β -SMAD³¹. RNA-seq results showed that *Wwtr1* was upregulated in 3D culture, in line with a unique ATAC-seq peak where lies a SMAD4 binding motif (GTCT) (**Fig. 1G**), suggesting that the chromatin was open to SMAD4 binding under normoxic 3D conditions. Interestingly, we did not observe significant changes of ATAC-seq peaks with HIF-1a and HIF-2a (EPAS1) binding motifs albeit the HIF-1b (ARNT) binding motif was greatly enriched under hypoxic 2D and normoxic 3D conditions (**Fig. 1-figure supplement 3**). Considering that ARNT could heterodimerize with other bHLH transcription factors, these data may suggest that the chromatin accessibility for HIF-1a and HIF-2a is poised to be open in response to hypoxic stress.

Cellular fitness genes under 21% and 1% oxygen in 2D and 3D culture conditions

Next, we sought to understand the cellular fitness in different culture conditions. Previous CRISPR screens under hypoxia were carried out for two weeks in K562 leukemia cells in suspension and 5 days for U2OS osteosarcoma cells in a monolayer^{12,13}. While some cells may not tolerate chronic hypoxia in 2D culture, the ABC-Myc cell lines such as NEJF10 showed a similar doubling time over 4 weeks under normoxia and 1% oxygen tension in monolayer culture (**Fig. 2-figure supplement 1A**), probably due to its greater dependence on glycolysis for ATP production (**Fig. 2-figure supplement 1B**). Similarly, NEJF10 cells grew in 3D culture continuously (**Fig. 2-figure supplement 1C**). To understand the cell fitness difference in normoxia, chronic hypoxia or 3D conditions, we generated a genome-wide mouse CRISPR knockout pooled library (Brie, lentiCRISPRv2), which includes 1000 control gRNAs and 78,637 gRNAs targeting 19,674 genes. Following a 36-hour puromycin selection after NEJF10 was transduced with the pooled library, cells were split into three different culture conditions: normoxia and 1% oxygen tension cultured as a monolayer and normoxic 3D spheroids (**Fig. 2A**, **Fig. 2-figure supplement 2**). Samples cultured as a monolayer were collected at different time points (days 1, 6, 9, 11, 17, 19 and 23 posttransduction for normoxia, days 4, 6, 11, 14, 17, 19 and 23 post-transduction for 1% oxygen), which allowed tracking of the gradual depletion or accumulation of gRNAs for a given gene KO in the two oxygen tensions. However, we only had a one-time point sample (day 28 post-infection) for 3D spheroids. The MAGeCK algorithm was used to identify differential fitness genes for each time point³². For monolayer culture, we defined the fitness genes as “negative selection” or “positive selection” if their depletion or accumulation appeared as significant in at least 4 different time points (Supplementary file 9). We identified 648 common genes in negative selection and 6 common genes in positive selection under all conditions. While the positive selection genes were basically tumor suppressor genes such as *Cdkn2a*, *Pten* and *Ambra1*, the negative selection genes engaged in essential biological processes including RNA polymerase, DNA replication, ribosome, spliceosome, and pathways such as MYC and E2F (**Fig. 2-figure supplement 3A, B**). Protein-protein interaction network analysis further demonstrated that the essential genes under all three conditions encode proteins involved in splicing, transcription, translation, proteasome, and other critical cellular functions (**Fig. 2-figure supplement 3C**).

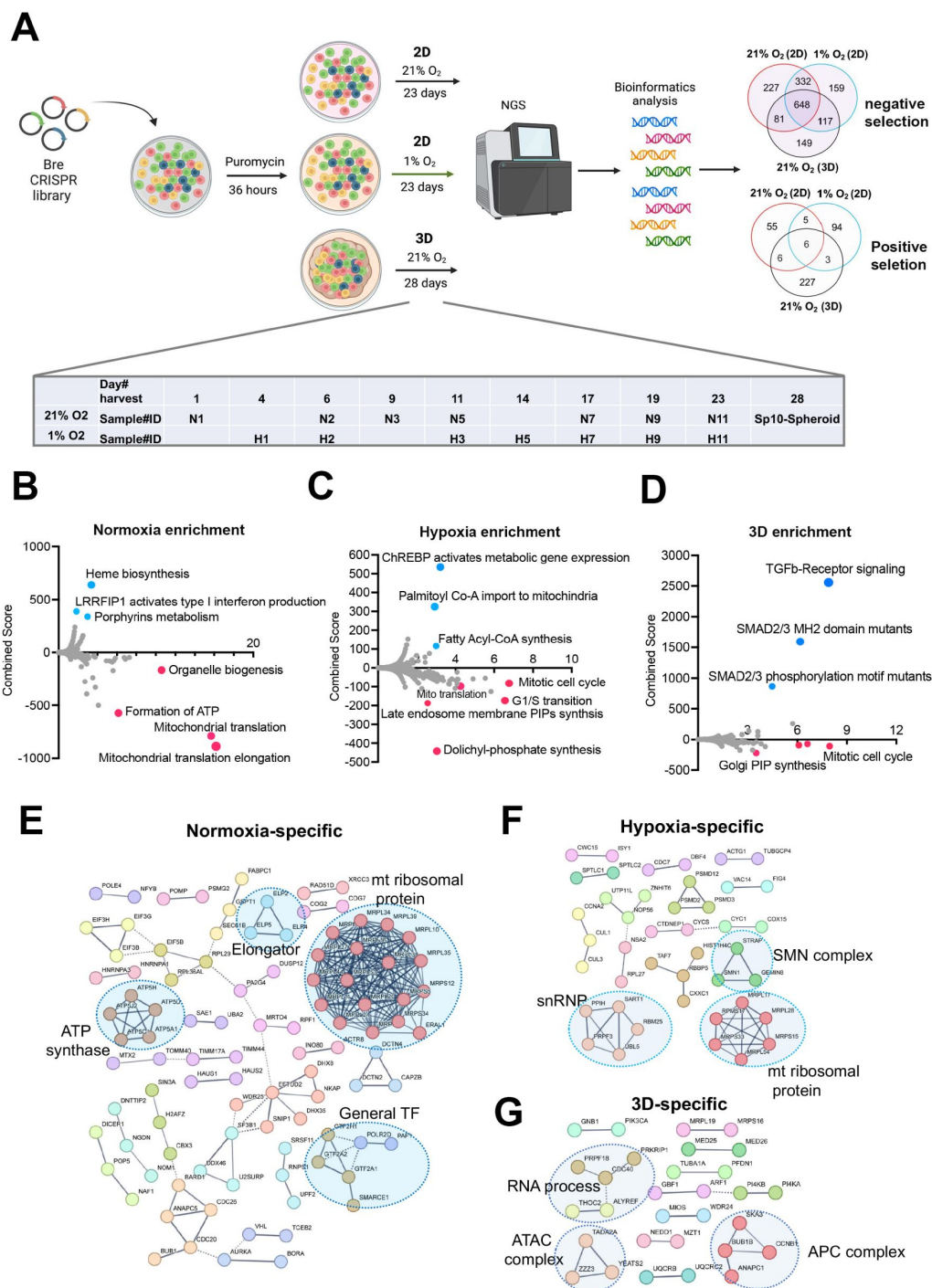


Figure 2.

Identification of cell fitness genes in 2D hypoxia, 2D normoxia, and 3D normoxia.

(A) Scheme showing the procedure of genome-wide CRISPR screen of the ABC-Myc NEJF10 cell line treated with different growth environments. Venn analysis of essential genes (negative selection) and anti-proliferative genes (positive selection) reveals gene essentiality for cellular fitness identified in 2D normoxia, 2D hypoxia, and 3D Normoxia. The table is a report of when samples were harvested with N representing 2D normoxia, H representing 2D hypoxia, and Sp10- Spheroid representing 3D normoxia grown cells.

(B, C, D) Pathway enrichment analysis reported as an enrichment score with negative values indicating negative selection, positive values indicating positive selection, and values of zero indicating no relationship of the respective condition. The colored points highlight significant upregulation (blue) or down regulation (red), whereas grey colored points were not significantly enriched.

(E, F, G) Pathways within a protein-protein interaction network of genes essential in cellular fitness are selectively enriched in normoxic, hypoxic, and 3D specific environments. Related genes are clustered by color with some circled to indicate functional complexes.

Since 3D culture itself leads to a relatively hypoxic core of cells within each spheroid and therefore partially recapitulates the transcriptional program of hypoxia pathway induction, we examined the fitness genes shared by monolayer under 1% oxygen tension and 3D under normoxia. Pathway analysis revealed that genes that engage in DNA repair and oxidative phosphorylation were essential to NEJF10 survival under both conditions (**Fig. 2-figure supplement 4A, B**). While hypoxia may promote a switch from oxidative phosphorylation to glycolysis, studies showed that the low oxygen concentrations in tumors may not be limiting for oxidative phosphorylation, and ATP is generated by oxidative phosphorylation in tumors even at very low oxygen tensions³³. Therefore, NEJF10 cells might be still dependent on oxidative phosphorylation for ATP production under hypoxia. Next, we examined the context-specific fitness genes. In monolayer culture under 21% oxygen tension, genes involved in mitochondrial translation, ATP production, and organelle biogenesis were enriched in the negative selection fraction, while genes involved in heme biosynthesis or porphyrin metabolism were enriched in positive selection (**Fig. 2B**). Heme is an essential molecule for living aerobic organisms. Heme biosynthesis involves an eight-step enzymatic pathway starting in mitochondria with the condensation of succinyl Co-A from the citric acid cycle and an amino acid glycine. Inactivating mutations in heme synthesis genes define a group of diseases known as porphyria³⁴. Recent studies revealed that acute hepatic porphyria is associated with increased risk of hepatocellular carcinoma (HCC)^{35, 36}. In addition, under 1% oxygen, cells in monolayer culture were more specifically sensitive to the loss of genes involved in cell cycle progression, dolichyl phosphate synthesis and to a subgroup of genes in mitochondrial ribosomal translation, while genes involved in lipid metabolism (ChREBP pathway, fatty acyl-coA synthesis) were enriched in positive selection (**Fig. 2C**). Interestingly, while some mitotic genes and Golgi phosphatidylinositol phosphate (PIP) synthesis genes were more essential to cells under 3D culture, the TGF β signaling pathway was uniquely and significantly enriched in positive selection in spheroids (**Fig. 2D**). Pan-cancer analysis reveals that genetic alterations in TGF β pathway members occurred in 39% of TCGA cases, which were correlated with the expression of metastasis genes and poor prognosis³⁷. Protein-protein interaction network analysis confirmed that mitochondrial translation genes, ATP synthase genes and general transcription factors were more specifically important for cell survival under 21% oxygen in 2D culture (**Fig. 2E**), while in 1% oxygen monolayer, RNA processing genes such as survival of motor neurons (SMN) complex and small nuclear ribonucleoprotein (snRNP) were more essential to cell survival (**Fig. 2F**). The SMN complex plays an essential role in the assembly of the spliceosomal snRNPs. One study has shown that neurons with low levels of SMN are more sensitive to hypoxia and will be the ones which undergo cell death first in any population³⁸, supporting the hypoxiaspecific role of SMN in our screening. However, in 3D culture under 21% oxygen, a subgroup of genes involved in RNA processing, the Anaphase-Promoting Complex (APC) for mitosis, and Ada Two A containing (ATAC) complex for acetylation of histone and Cyclin A/Cdk2³⁹, were important for spheroid growth (**Fig. 2G**). As most prior knowledge about mammalian cell cycle progression comes from 2D cell culture experiments, it remains to be understood how cells in 3D culture proceed from G1 to S, then from S to G2/M phase.

Cell fitness differences with VHL-HIF-1a pathway depletion under normoxia vs. 1% oxygen in 2D or 3D

Considering that the VHL-HIF-1a pathway plays a central role in response to hypoxic stress (**Fig. 3A**), it would be expected that knockout either of VHL or HIF-1a could change the cell fitness under normoxia and hypoxia. However, the two previous screenings of K562 and U2OS cells did not show a significant effect on cell fitness in either normoxia or hypoxia after deletion of HIF-1a^{12,13}. Rather, deletion of *VHL* in K562 cells led to a growth advantage under normoxia¹¹. In our model system, knockout of *Vhl* in NEJF10 cells led to gradual gRNA depletion in normoxia but no depletion of its gRNA under 1% oxygen (**Fig. 3B**), indicating that VHL loss is incompatible with cellular fitness under normoxic conditions. Conversely, knockout of *Hif1a* or *Arnt* (encoding HIF-1b) but not *Epas1* (encoding HIF-2a) led to gRNA depletion under 1% oxygen tension but not normoxia (**Figs. 3C-E**). We observed similar effects of VHL-HIF pathway inactivation in 3D conditions in which cells cannot tolerate the genetic deletion of *Hif1a* and *Arnt* (**Fig. 3F**). Cells in 3D even had a remarkable increase in gRNA counts of the *Vhl* gene (**Fig. 3F**). By examining the DepMAP data, we found that nearly all human cancer cells cultured under 2D normoxia cannot tolerate the loss of *VHL*, while cells tended to gain growth benefit by *HIF1A* knockout under normoxic oxygen conditions (**Fig. 3G**). Analysis of DepMAP data revealed that the *VHL* knockout effect was significantly negatively correlated with *HIF1A* expression levels under normoxia but had a less significant correlation with *HIF2A* and *HIF1B* (**Figs. 3H**, **Fig. 3-figure supplement 1**). We further validated the loss of function of VHL in HepG2, a human liver cancer cell line. We were unable to obtain a complete knockout of VHL in HepG2 cells when we generated stable clones that were cultured regularly in 2D under 21% oxygen (**Fig. 3I**), suggesting VHL is essential to cell survival under this culture condition. Partial loss of VHL led to a significant reduction in cell proliferation in 2D but not in 3D culture under 21% oxygen tension (**Figs. 3J, K**), which is in line with our screening result in NEJF10 cells. Thus, these genetic data indicate that VHL-HIF1 plays a dominant role in determining the cell fitness in normoxia and hypoxia, although cell type specific effects for deletion of VHL-HIF1 may also exist. Nevertheless, why some cells like U2OS were less sensitive to loss of HIF1 remains to be studied. The opposite phenotype of VHL deletion in K562 and NEJF10 cells under 21% oxygen indicates that cell type-specific functions of VHL may be present.

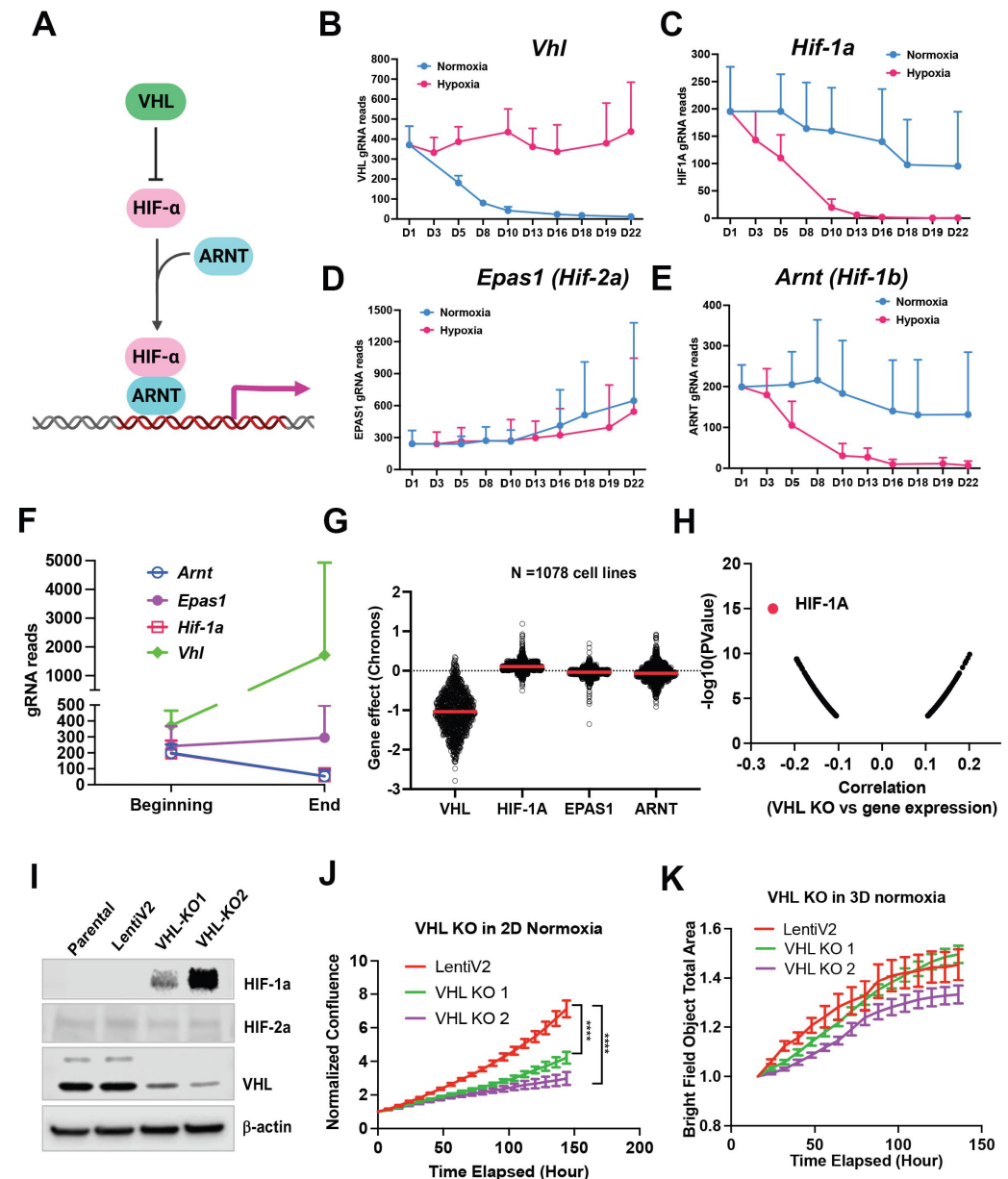


Figure 3.

Fitness incompatibility of VHL-HIF1a pathway in normoxia vs hypoxia or 3D.

(A) Scheme representing the VHL-HIF pathway. The HIF-1 and HIF-2a family of transcription factors are degraded by VHL, an E3 ubiquitin ligase, in normoxia. In the presence of hypoxia or loss function of VHL, HIFa is present and dimerizes with nuclear HIF1b or ARNT to drive gene transcription.

(B-E) The gRNA reads for *Vhl*(B), *Hif-1a* (C), *Hif-2a* or *Epas1* (D), *Hif-1b* or *Arnt*(E) at different time points in 2D normoxia and 2D hypoxia. Data = mean $\pm$ SD (n =4 for each time point)

(F) The gRNA reads for *Vhl*, *Hif-1a*, *Hif-2a* or *Epas1*, *Hif-1b* or *Arnt* at the beginning and end time points in 3D normoxia.

(G) CRISPR KO effect of *VHL*, *HIF-1A*, *HIF-2A* (*EPAS1*), *HIF-1B* (*ARNT*) in 1078 human cell lines. Data were extracted from DepMAP database (www.depmap.org).

(H) Spearman correlation of *VHL* CRISPR KO effect vs *HIF-1A* gene expression in 1078 human cell lines.

(I) Western blot analysis of *VHL* CRISPR knockout in HepG2 cells using two gRNAs with indicated antibodies.

(J) Cell growth of wild-type and *VHL* KO HepG2 over time monitored by Incucyte live cell microscopy in real-time in 2D normoxic condition. **** $P < 0.0001$. p value is calculated by student t test for the last time point data. Data = mean $\pm$ SD.

(K) Cell growth of wild-type and *VHL* KO HepG2 over time monitored by Incucyte in real-time in 3D normoxic condition. Data = mean $\pm$ SD.

Distinct fitness outcomes for respiratory chain complex loss of function

Since mitochondrial translation and ATP synthase genes seemed to be essential for cell survival under normoxia cultured in 2D, we compared the time course of gRNA enrichment for genes with mitochondrial function. In comparison with gRNA reads on day 1 before splitting cells into different culture conditions, there was a significant reduction of gRNA counts on day 5 under normoxia for the mitochondrial ribosomal genes (**Figs. 4A, B**), in contrast to 1% oxygen which showed no difference between day 5 and day 1. Even on day 10, gRNA counts for mitochondrial translation genes were relatively more abundant in 1% oxygen than normoxia ($p = 0.085$) (**Fig. 4B**), indicating that hypoxia buffers the blockade effect of mitochondrial protein translation. The large-scale CRISPR screening of 1095 cell lines under normoxia performed by the DepMAP project showed that the knockout effect of human mitochondrial ribosomal gene *MPRS22* (as one example) was correlated with *HIF1A* expression (**Fig. 4C**), supporting the hypothesis that cells may survive longer under hypoxia when mitochondrial translation is inhibited. The electron transport chain is composed of 5 complexes (complex I-V), which drives ATP production by relaying electrons to oxygen. Surprisingly, the knockout of each complex gave rise to distinct fitness outcomes under normoxia and 1% oxygen tensions in monolayer and 3D culture. While knockout of complex I and IV led to a cell fitness advantage in all three conditions (which was opposite to K562 and U2OS cells which cannot tolerate complex I loss under normoxia^{12,13}), loss of function of complex II and III seemed to be detrimental to NEJF10 cells in both 21% and 1% oxygen in monolayer, although gRNA counts were relatively enriched in 3D (**Fig. 4D**). However, genetic deletion of complex V, the ATP synthase, led to an adverse effect on cell survival under normoxia, opposite to 1% oxygen conditions in which cells gained a fitness advantage with knockout of complex V (**Fig. 4D**). For example, gRNAs for the *Atp5c1* gene, encoding one key component of ATP synthase, were gradually depleted under 21% oxygen but increased under 1% oxygen over time (**Fig. 4E**). The large-scale CRISPR screening of 1095 cell lines under normoxia by the DepMAP project showed that human *ATP5C1* is a pan-essential gene (**Fig. 4F**) because nearly all cells cannot tolerate the loss of *ATP5C1*. There was a negative correlation between *ATPC1* knockout and *HIF1A* expression (**Fig. 4G**). Because *HIF1A* can serve as a hypoxia surrogate marker, this negative correlation independently validated our observation that cells with loss of the ATPase survive better in hypoxia than normoxia.

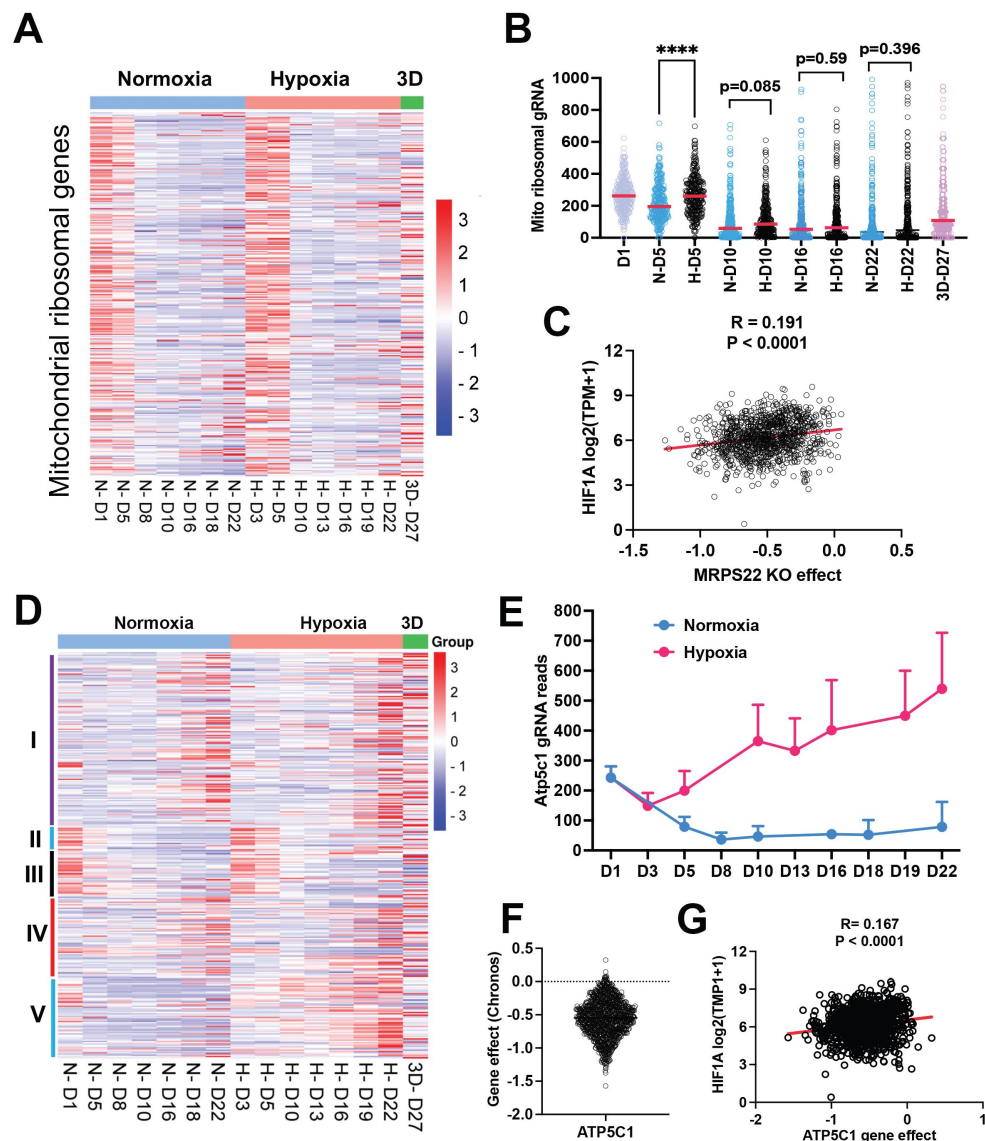


Figure 4.

The effect of mitochondrial compartments on cell fitness.

(A) Heatmap showing the gRNA reads for mitochondrial ribosomal genes at different time points in 2D normoxia, 2D hypoxia and 3D normoxia. Scale bar indicates Z score.

(B) Comparison of gRNA reads for mitochondrial ribosomal genes at different time points in 2D normoxia, 2D hypoxia and 3D normoxia. ****p<0.0001. p value is calculated by student t test for the last time point data. Data = mean±SD.

(C) Spearman correlation of *MRPS22* CRISPR KO effect vs *HIF-1A* gene expression in 1078 human cell lines.

(D) Heatmap showing the gRNA reads for mitochondrial electron transport genes at different time points in 2D normoxia, 2D hypoxia and 3D normoxia. Scale bar indicates Z score.

(E) The gRNA reads for *Atp5c1* at different time points in 2D normoxia and 2D hypoxia. Data = mean $\pm$ SD

(F) CRISPR KO effect of *ATP5C1* in 1078 human cell lines. Data were extracted from DepMAP database (www.depmap.org).

(G) Spearman correlation of *ATP5C1* CRISPR KO effect vs *HIF-1A* gene expression in 1078 human cell lines.

Therapeutics targeting the oxidative phosphorylation (OXPHOS) pathway are being evaluated in clinical trials. In two recent phase one clinical trials for treatment of advanced solid cancers and acute myeloid leukemia, a complex I inhibitor IACS-010759 showed limited antitumor activity⁴⁰. While in-depth study is needed to understand the mechanism by which IACS-010759 failed in clinic, the anticancer activity of IACS-010759 seemed to be inversely correlated with *HIF1A* expression (Fig. 4-figure supplement 1). Therefore, the anticancer activity of IACS-010759 might be greatly blunted by hypoxic conditions in a tumor. Considering cancer cells such as NEJF10 could gain a fitness advantage when complex I is genetically inhibited, pharmacological inhibition of complex I could even promote tumor growth under some specific settings.

Epigenetics and organogenesis pathways function as important fitness mechanisms in 3D culture

There were fewer genes which were enriched in the CRISPR screen whose loss of function promoted cell fitness under the three conditions. Protein interaction network analysis further validated context-specific dependencies such as heme biosynthesis genes (*Alas1*, *Hmbs*) under 21% oxygen in 2D culture, and fatty acid synthesis and mitochondrial respiratory-chain complex (MITRAC) genes (*Cox18*, *Surf1*, *Tmem70*) under 1% oxygen in 2D culture (Figs. 5A, B). Under 3D culture conditions, protein interaction networks were enriched with genes involved in epigenetics (*Kmt2d*, *Bcor*, *Arid2*, *Phf21b*, etc), TGF β -SMAD signaling pathway (*Tgfb1*, *Tgfb2*, *Tgfb3*, *Smad3*, *Smad4*, *Runx3*), Hippo signaling pathway (*Nf2*, *Lats1*, *Sav1*, *Frmd6*), and M6A methyltransferase complex (*Mettl3*, *Mettl14*, *Wtap*) (Fig. 5C). After carefully examining the enriched genes based on literature annotation, we found that the signaling pathways involved in multicellular organogenesis such as TGF β , Wnt, Hedgehog and Notch pathways were enriched, particularly in 3D culture (Fig. 5D). This finding supports the theory that breakdown of communication and coordination between cells (which is required for multicellularity) leads to tumorigenesis and cancer progression^{41–43}. TGF β -SMAD2/3/4 pathway is known to be involved in tumorigenesis. Interestingly, the BMP-SMAD1/5/8 signaling was not enriched in the screenings, indicating the TGF β -SMAD2/3/4 pathway is particularly important for restricting 3D cell proliferation. Considering that the aforementioned organogenesis signaling pathways converge on nuclear gene transcription, the above-mentioned epigenetic modifiers are highly likely to be integrated into these signaling pathways and modulate gene transcription. Notably, most of these genes (*Kmt2d*, *Bcor*, *Smad*, *Nf2*, *Lats1*) are tumor suppressors in human cancers. KMT2D (also known as MLL4) is a methyltransferase that methylates H3K4, and which frequently exhibits loss of function mutations in a variety of human cancers⁴⁴ (Fig. 5-figure supplement 1A). Interestingly, DepMAP CRISPR screens showed that *KMT2D* was a pan-essential gene in 2D normoxia regardless of its mutation status (Fig. 5-figure supplement 1B). This was consistent with our results which showed *Kmt2d* gRNA depletion under both 21% and 1% oxygen tensions in 2D culture, and which contrasts with conventional understanding that cells should gain a proliferation advantage after knockout of a tumor suppressor gene. However, under 3D culture, *Kmt2d* gRNAs were remarkably enriched (Fig. 5E), which supports the tumor suppressor role of *KMT2D*. *BCOR* is another frequently mutated tumor suppressor gene⁴⁵. Knockout of *Bcor* in NEJF10 cells promoted cell fitness in hypoxia and more dramatically in 3D but had minimal effect in normoxia (Fig. 5E). DepMAP data showed that the average effect of *BCOR* gene knockout was close to net zero (Fig. 5-figure supplement 2A). These data underscore that careful interpretation of a CRISPR knockout phenotype for tumor suppressor genes needs to consider the cellular context. *Wtap*, *Mettl3*, and

Mettl14 encode proteins forming the WMM complex that acts as a N6-methyltransferase to methylate adenosine residues at the N(6) position of some mRNAs (m⁶A) to regulate mRNA stability. Similar to the fitness outcomes of *Kmt2d* in the three conditions, knockout of *Wtap*, *Mettl3* and *Mettl14* led to great gRNA count enrichment in 3D culture but reduction in 2D culture (**Fig. 5E** [↗](#)). The 3D-specific role of the WMM complex was further verified in NEJF6 cells in which gRNA reads of WMM complex were remarkably increased under 3D culture as in NEJF10 cells (**Fig. 5F** [↗](#), Supplementary file 10). DepMAP data showed that *WTAP*, *METTL3* and *METTL14* are commonly essential to most if not all cells (**Fig. 5-figure supplement 2A** [↗](#)). While most previous studies recognize that *METTL3* and *METTL14* are important to cancer cell survival, one study showed that *METTL3* enhanced the tumor suppression activity of p53⁴⁶ [↗](#). A recent study reported that knockout of *Mettl3* enhances liver tumorigenesis in multiple mouse models⁴⁷ [↗](#). Two studies also indicate that *METTL14* acts as a tumor suppressor by facilitating DNA repair or modulating glycolysis^{48,49} [↗](#). By analyzing DepMAP data, we further confirmed the functional connection of the WMM complex with p53. The knockout effect of *METTL3* was negatively correlated with the presence of *TP53* mutations (**Fig. 5G** [↗](#)), positively correlated with the knockout effect of *TP53*, but negatively correlated with the knockout of *MDM2* (**Fig. 5-figure supplement 2B, C** [↗](#)). In line with these observations, the knockout effect of *METTL3* was positively correlated with *MDM2* expression (**Fig. 5-figure supplement 2D** [↗](#)). *MDM2* is known to antagonize the functions of p53. Thus, despite the overall detrimental effect of WMM knockout under 2D normoxia conditions from the DepMAP data, we were able to establish the genetic connection of WMM-p53-MDM2, supporting the observation WMM is tumor suppressive. It is important to note that we cannot exclude the possibility that these genes may have cancer-specific functions, and may behave differently in different cancer lineages. Taken together, our data indicate that organogenesis signaling and epigenetic regulators could behave drastically differently in 2D vs 3D culture conditions.

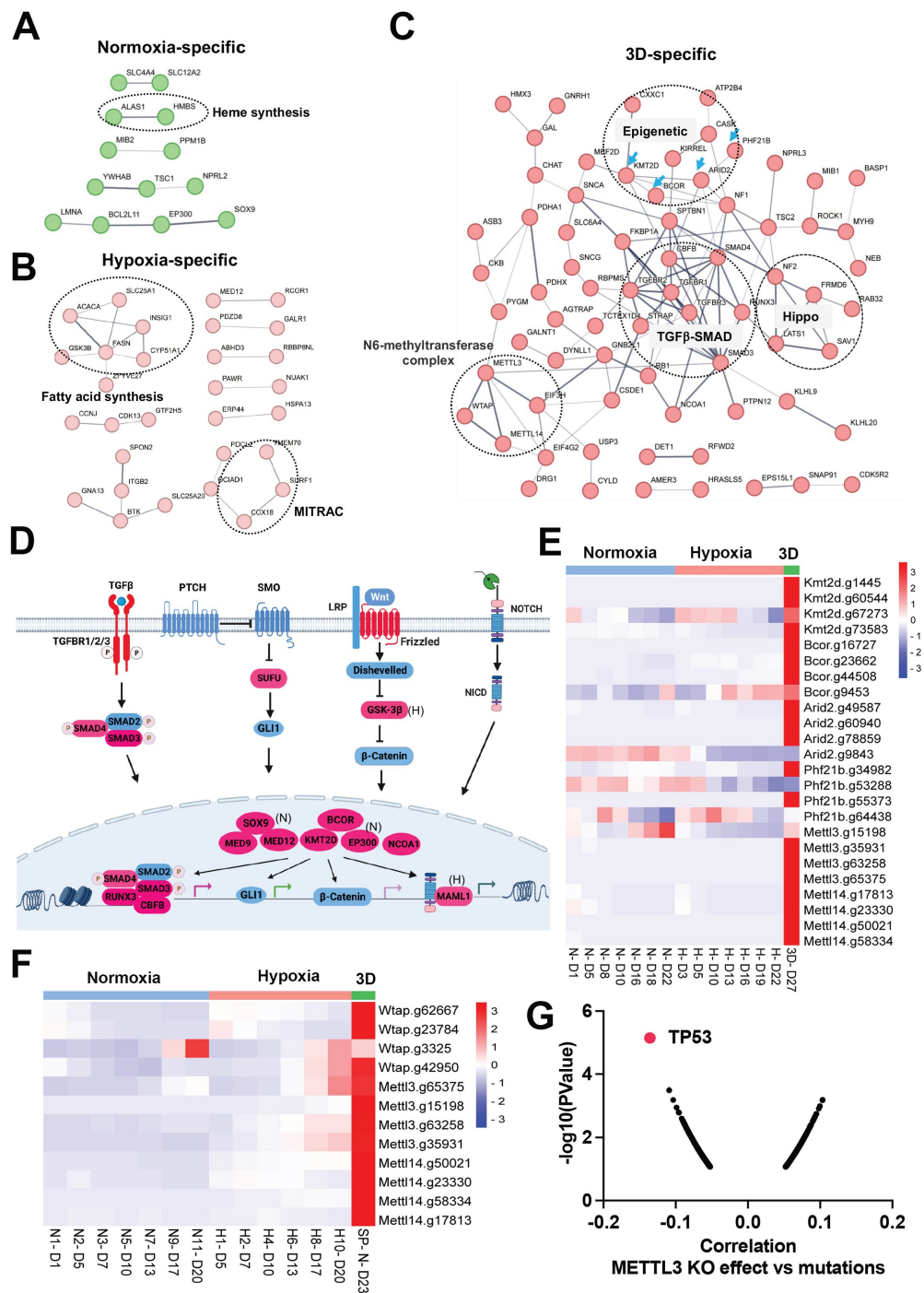


Figure 5.

Organogenesis signaling and epigenetic modifiers constrain 3D cell proliferation

(A-C) Positive selection of pathways within protein-protein interaction networks enriched specifically in 2D normoxia, 2D hypoxia, and 3D normoxia.

(D) The graphic illustrates organogenesis signaling pathways positively selected in 3D. Please note that we included hits from normoxia (Sox9, Ep300) and hypoxia (Mam1 and Gsk-3b) in the related pathways.

(E) The heatmap for the normalized gRNA reads for epigenetic genes in 2D hypoxia and normoxia and 3D normoxia. Red color indicates positive selection. Scale bar indicates Z score.

(F) The heatmap for the normalized gRNA reads for WMM complex of NEJF6 cells in 2D hypoxia and normoxia and 3D normoxia. Scale bar indicates Z score.

(G) Spearman correlation of METTL3 CRISPR KO effect vs TP53 mutations in human cancer cells lines by analyzing DepMAP data (www.depmap.org).

Context-specific fitness genes with altered gene expression in hypoxia and 3D are enriched with metabolic pathways

Two previous studies showed that HIF targets were not enriched in identified fitness genes under hypoxia^{12,13}. HIF mainly upregulates gene expression and the downregulated genes caused by hypoxia were not directly induced by HIF⁵⁰. In the current study, pathway enrichment analysis showed that the hypoxia-downregulated genes were negatively selected by CRISPR knockout (**Fig. 6-figure supplement 1A**). These negative fitness genes were enriched in the pathways of MYC, Wnt-b-Catenin and NOTCH4, as well as mitochondrial metabolism (**Fig. 6-figure supplement 1A**). However, among the positive selection fitness genes whose expression was altered by hypoxia, lipid metabolism genes were most significantly enriched (**Fig. 6-figure supplement 1B**). Among the genes whose expression was altered by 3D culture, the hypoxia-downregulated genes were also negatively selected, probably due to induction of the hypoxia pathway by 3D culture (**Fig. 6-figure supplement 1C**). Different from hypoxia, however, the 3D genes in negative selection were enriched in pathways of MYCN, BMP2, DREAM complex and genetic modifiers such as DNA and histone methylases, and histone acetyltransferases (**Fig. 6-figure supplement 1C**). For the 3D genes positively selected in the screening, the TGFb-SMAD pathway and NOTCH pathway were remarkably enriched (**Fig. 6-figure supplement 1D**), further suggesting that loss of genes in multicellular communication promotes tumor progression.

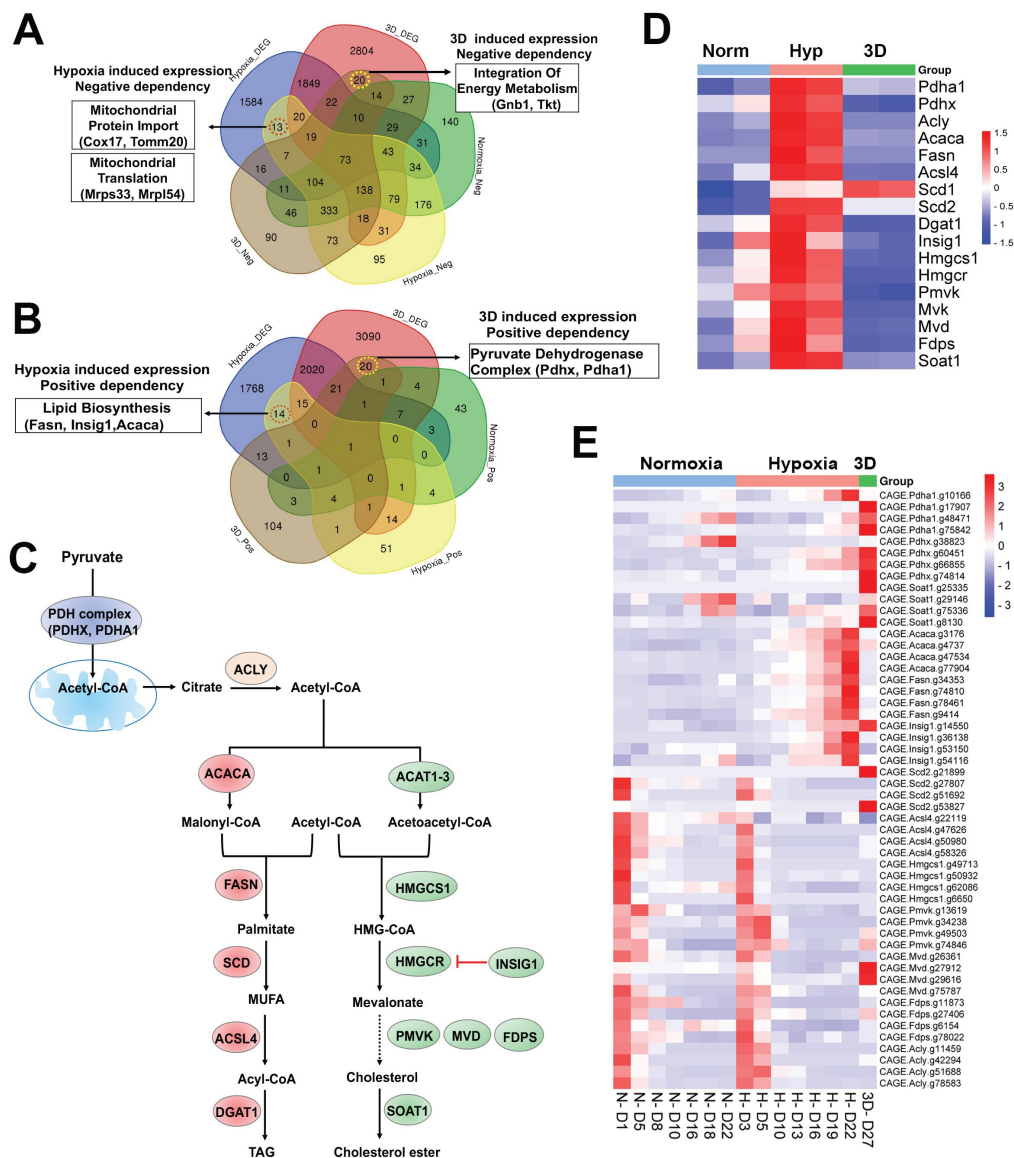


Figure 6.

Selective fitness of lipid metabolisms.

(A) Venn diagram showing 13 and 20 genes with negative dependency in specifically hypoxia and 3D induced conditions, respectively, are selectively induced in hypoxia or 3D.

(B) Venn diagram showing 14 and 20 genes with positive dependency in specifically hypoxia and 3D induced conditions, respectively, are selectively induced in hypoxia or 3D.

(C) A diagram showing lipid biosynthesis pathway of unsaturated fatty acids and cholesterol.

(D) The heatmap showing the gene expression involved in lipid biosynthesis. Scale bar indicates Z score.

(E) The heatmap showing the normalized gRNA reads for genes involved in lipid biosynthesis in 2D normoxia and hypoxia, 3D normoxia. Scale bar indicates Z score.

Next, we examined context-specific fitness genes whose expression and CRISPR gRNAs were selectively altered in 1% oxygen or 3D conditions. We only found 13 and 14 genes in negative and positive selection screenings respectively under 1% oxygen, 20 and 20 genes in negative and positive selection screenings respectively in 3D culture (**Figs. 6A, B** [↗](#), Supplementary file 11). Several hypoxia-inducible genes involved in mitochondrial import (*Cox17* and *Tomm20*) and mitochondrial translation (*Mrps34*, *Mrpl54*) were selectively essential to cell fitness under 1% oxygen tension (**Fig. 6A** [↗](#)). *Pdgfrb*, another hypoxia-inducible gene involved in cell proliferation and angiogenesis, was also an important fitness gene in 1% oxygen. However, hypoxia-inducible genes involved in regulation of fatty acids and cholesterol (*Fasn*, *Insig1*, *Acaca*) were positively selected under 1% oxygen tension (**Figs. 6B-E** [↗](#)). ACACA and FASN are responsible for the de novo biosynthesis of long-chain saturated fatty acids starting from acetyl-CoA and malonyl-CoA in the presence of NADPH, while INSIG1 negatively regulates HMGCR for inhibition of cholesterol synthesis (**Fig. 6C** [↗](#)). The selective 3D inducible genes involved in integration of energy metabolism (*Gnb1*, *Tkt*) were particularly essential to cells in 3D culture (**Fig. 6A** [↗](#)), while those involved in regulation of the pyruvate dehydrogenase (PDH) complex (*Pdhx*, *Pdha1*) were positively selected after knockout in 3D culture albeit to a lesser degree under 1% oxygen (**Figs. 6B, 6E** [↗](#)). GNB1 is a guanine nucleotide-binding protein (G protein) involved as a transducer in transmembrane signaling, and whose gain of function mutations promote myeloid transformation⁵¹ [↗](#). TKT is a thiamine-dependent enzyme which plays a role in the pentose phosphate pathway. The PDH complex is known to convert pyruvate to acetyl-CoA for citrate synthesis in the tricarboxylic acid cycle and cholesterol and fatty acid synthesis (**Fig. 6C** [↗](#)). It is likely that blockade of acetyl-CoA production by PDH knockout may force cells to use alternative energy sources under hypoxic and 3D conditions, averting the Warburg effect and promoting cell survival under limited oxygen and nutrient availability in 3D spheroids. The CRISPR results were corroborated by the findings that the genes involved in regulation of PDH complex were selectively upregulated in normoxic 3D conditions (**Fig. 1A** [↗](#)). This hypothesis awaits further validation in future studies. It is noteworthy that the activity of PDH is regulated by pyruvate dehydrogenase kinase (PDK). *Pdk1* underwent distinct alternative splicing changes in hypoxia and 3D (**Fig. 1-figure supplement 1E** [↗](#)), which may consequently affect the activity of PDH.

Distinct dependency of fatty acid and cholesterol synthesis pathways

Previous studies identified lipid metabolic genes which are critical to cell fitness under hypoxia¹³ [↗](#). However, lipid metabolic genes showed a more complex fitness phenotype although their expression was induced by hypoxia (**Fig. 6D** [↗](#)). For example, blockade of saturated fatty acid synthesis by knockout of *Fasn* and *Acaca* promoted cell fitness under hypoxia, while *Scd2*, encoding a stearoyl-CoA desaturase that utilizes oxygen and electrons from reduced cytochrome b5 to introduce the first double bond into saturated fatty acyl-CoA substrates, was essential to NEJF10 cells in both normoxia and hypoxia (**Fig. 6E** [↗](#)). This was distinct from the K562 cells which were selectively sensitive to *SCD* (human homolog of *Scd2*) KO in hypoxia¹³ [↗](#). *Acsl4*, encoding acyl-CoA synthetase long chain family member 4 protein that was selectively essential in hypoxia to K562 cells¹³ [↗](#), was essential to NEJF10 cells in all culture conditions (**Fig. 6E** [↗](#)). While the peroxisome was reported to be critical to K562 cell survival grown in hypoxia¹³ [↗](#), NEJF10 cells were not as sensitive to peroxisome loss. Analysis of DepMAP data revealed that the knockout effect of peroxisome genes (i.e., *PEX1*) had no significant correlation with the expression of either *HIF1A* or *MYC* while *SCD* knockout effect was negatively correlated with *MYC* expression (**Fig. 6-figure supplement 2A-D** [↗](#)), suggesting that *SCD* is a *MYC* cancer dependency gene. However, *SCD* knockout effect was positively correlated with *HIF1A* expression. Similar to *SCD*, *ACSL4* knockout effect was also negatively correlated with *MYC* expression but positively correlated with *HIF1A* expression (**Fig. 6-figure supplement 2E, F** [↗](#)). The effect of the *SCD* inhibitors A-939572 and MK-8245 was negatively impacted by high *ASCL4* expression (**Fig. 6-figure supplement 2G, H** [↗](#)), in line with the fact that *ASCL4* is downstream of *SCD* (**Fig. 6C** [↗](#)). These data further support that *SCD* and *ASCL4* are posited in one metabolic pathway in regulating cell fitness. Taken together, our

data suggest that saturated and non-saturated fatty acid synthesis exert opposite functions in regulating cell fitness, at least in NEJF10 cells. The ratio of saturated vs unsaturated fatty acids is critical to cell survival⁵². This possibly because saturated fatty acids are toxic to cells due to inducing endoplasmic reticulum (ER) stress⁵². Hypoxia induces high expression of *Acaca* and *Fasn* in NEJF10 cells indicating that hypoxia promotes saturated fatty acid synthesis, which is in line with the observation by Jain¹¹. The beneficial effect of *Fasn* and *Acaca* KO to NEJF10 under hypoxia is probably due to reduction of saturated fatty acid synthesis, and this hypothesis needs to be tested in the future. Although *Scd2* in NEJF10 cells was induced by hypoxia, it is possible that this is a compensatory induction because SCD proteins need oxygen for their activity. Thus, cells may be particularly sensitive to inhibition of stearoyl-CoA desaturase under hypoxic conditions.

The cholesterol biosynthesis pathway, another downstream metabolic branch of acetyl-CoA (Fig. 6C), was essential to cell survival since under either culture condition cells cannot survive after knockout of *Hmgcs1*, which encodes 3-hydroxy-3-methylglutaryl-CoA synthase 1, an enzyme catalyzing the condensation of acetyl-CoA with acetoacetyl-CoA to form HMG-CoA for cholesterol synthesis. The downstream enzymes for cholesterol synthesis such as *Pmvk*, *Mvd*, *Fdps* were essential for cell fitness under both normoxia and hypoxia in monolayer culture (Fig. 6E). Considering that cholesterol is required for membrane biogenesis and maintains the integrity and fluidity of cell membranes, cancer cells may not survive when the cholesterol synthesis pathway is fully shut down. MYC-driven cancers may be particularly sensitive to interruption of cholesterol synthesis since MYC is linked to dysregulation of cholesterol transport and storage⁵³. A previous study has shown that inhibition of cholesterol synthesis by statins prevents and reverses MYC-induced lymphomagenesis⁵⁴. Therefore, targeting cholesterol synthesis might be an option for MYC-driven cancers.

Synthetic lethality of partial loss of PRMT5 under 3D but not 2D culture

The spliceosome is essential to cell survival regardless of culture conditions (Fig. 2-figure supplement 1C). One of these key essential genes is *Prmt5* (Fig. 7A), which encodes splicing factor PRMT5. PRMT5 is required for survival of MYC-driven cancer cells⁵⁵, and has been extensively studied as a potential cancer therapeutic target. The essentiality of PRMT5 was further validated in another independent CRISPR screen in the NEJF6 cell line (Supplementary file 10), another MYC-driven liver cancer cell line derived from ABC-Myc mouse liver tumor²⁶. Interestingly, shRNA-mediated knockdown of *Prmt5* significantly reduced the growth of NEJF10 spheroids while minimal effect was seen when NEJF10 was cultured in monolayer under 21% and 1% oxygen tension (Fig. 7B-E). As shRNAs usually partially deplete the expression of target genes as evidenced by the western blot analysis, these data indicate that complete loss of *Prmt5* is needed to inhibit NEJF10 proliferation under 2D culture conditions. We further validated the role of *Prmt5* by genetically deleting *Prmt5* from liver via breeding a floxed *Prmt5* mouse strain with the ABC-Myc mice²⁶. In comparison with the ABC-Myc control mice that rapidly developed liver tumors, knockout of either one allele or both alleles of *Prmt5* extended comparable mouse survival (Fig. 7F). H&E staining of the liver tissues revealed that deletion of even one *Prmt5* allele led to massive necrosis in some tumor regions with inflammatory cell infiltration (as evidenced by macrophage surrounding of the necrotic areas). This was not observed in the control mice (Fig. 7G, Supplementary file 12). These data verified the importance of *Prmt5* for tumor cell survival under multicellular settings, and demonstrate even partial loss of *Prmt5* function *in vivo* may lead to a comparable effect to complete *Prmt5* loss. Notably, the control mouse livers with *Prmt5* knockout appeared to be normal, indicating that *Prmt5* is essential for the MYC-transformed cancer cells but not for the non-transformed liver cells.

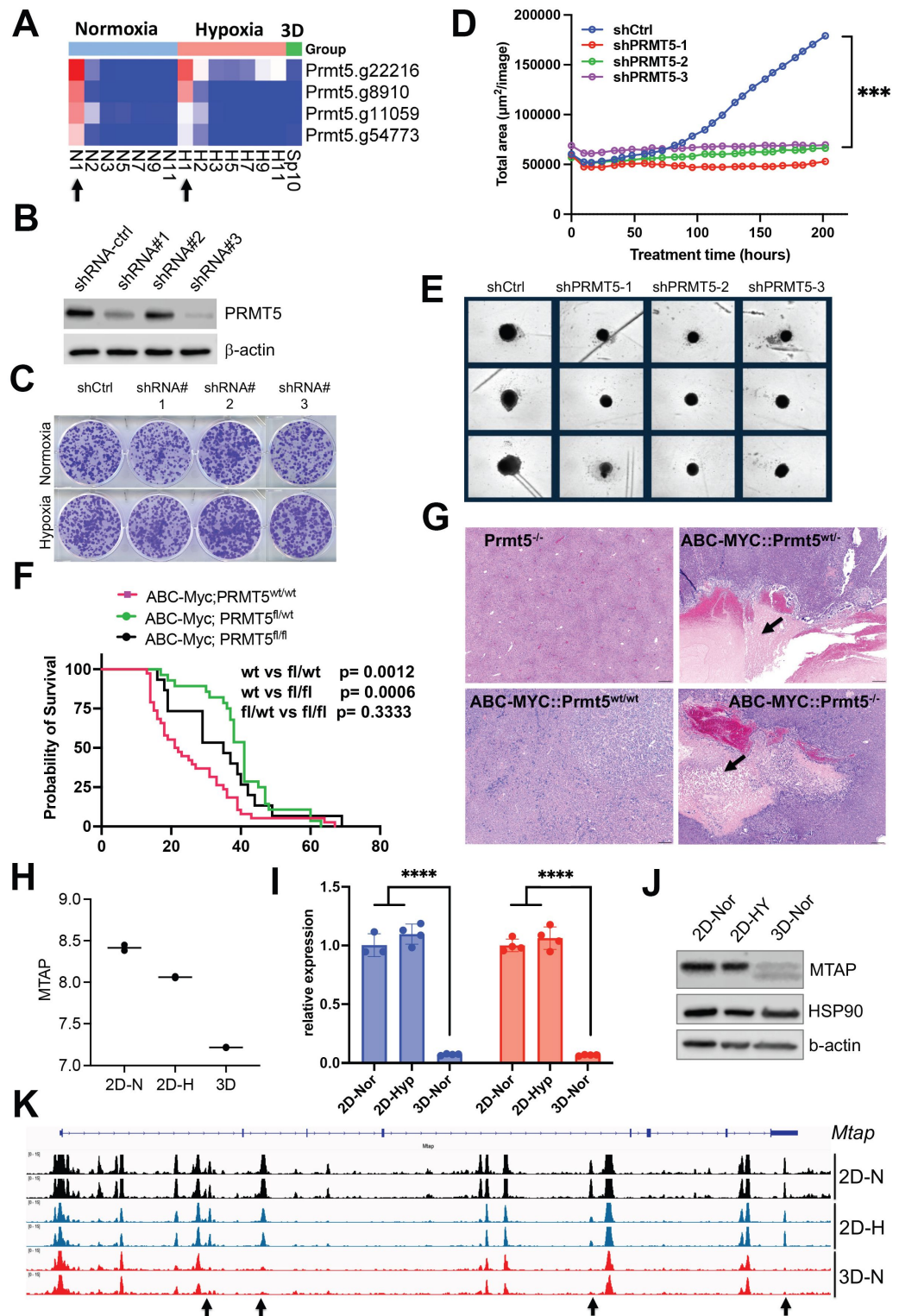


Figure 7.

Synthetic lethality of partial *Prmt5* loss with 3D.

- (A) The heatmap showing the normalized gRNA reads for *Prmt5* in 2D normoxia and hypoxia, 3D normoxia. Arrows indicate the first samples harvested for analysis.
- (B) Western blot analysis with indicated antibodies after *Prmt5* knockdown in NEJF10 cells.
- (C) Colony formation of NEJF10 cells after 5 days of *Prmt5* knockdown in 2D normoxia and hypoxia. Cells were stained with crystal violet.
- (D) Incucyte monitoring of cell proliferation grown in 3D after *Prmt5* knockdown in NEJF10 cells. *** $p < 0.001$ for comparison of shCtrl with each of shPrmt5. p value is calculated by student t test by comparing the last reading. N = 3 per group. Data = mean $\pm$ SD.
- (E) Snapshot of NEJF10 cell with or without *Prmt5* knockdown in 3D.
- (F) Kaplan-Meier survival for mice of ABC-Myc::Prmt5^{+/+}, ABC-Myc::Prmt5^{+/-}, and ABC-Myc::Prmt5^{-/-}. P values are calculated by log-rank test.
- (G) Hematoxylin and eosin stain for tumor tissues obtained from mice of Prmt5^{-/-}, ABC-Myc::Prmt5^{+/+}, ABC-Myc::Prmt5^{+/-}, and ABC-Myc::Prmt5^{-/-}. Arrows indicate necrosis in tumor areas. Scale bar = 500μm.
- (H) Quantification of normalized *Mtap* mRNA under 2D normoxia, 2D hypoxia and 3D normoxia from RNA-seq results. n=2 for each group.
- (I) Real-time quantitative PCR to determine the expression of *Mtap* in NEJF10 cells cultured under 3D normoxia, 2D normoxia and hypoxia for 3 days. Blue and orange indicate results obtained from two different pairs of PCR primers against *Mtap*. n=3 for each group. **** $p < 0.0001$ student t test.
- (J) Western blot analysis with indicated antibodies of NEJF10 whole cell lysates cultured from 3D normoxia, 2D normoxia and hypoxia for 3 days.
- (K) IGV snapshot showing the ATAC-seq result for *Mtap* gene. The black arrow indicates the enhancers in *Mtap* gene are selectively lost in 3D.

Epigenetic reprogramming in 3D culture leads to downregulation of *Mtap*

We sought to understand why partial loss of *Prmt5* affected cell proliferation in 3D but not in 2D culture. It is well known that the *MTAP* (encoding 5-methylthioadenosine phosphorylase) deletion is synthetically lethal to genetic ablation of *PRMT5*⁵⁷, and this finding has been translated to clinical trials by using PRMT5 inhibitors for tumors with *MTAP* deletions. Here we further verified the expression of *MTAP* and *PRMT5* knockdown effect (Fig. 7-figure supplement 1). Since we have observed unique pre-mRNA splicing changes in 3D culture, we examined if *Mtap* underwent distinct splicing. Indeed, in comparison with the 2D culture which showed a similar splicing pattern in normoxia and hypoxia, *Mtap* displayed distinct exon skipping of exon 2 and/or 3 in 3D culture (Fig. 7-figure supplement 2), which may potentially hinder its enzymatic activity by abrogating sulfate/phosphate binding⁵⁷. However, only a small fraction of *Mtap* pre-mRNA transcripts underwent these events. Notably, the exon junction reads of whole *Mtap* pre-mRNA transcript were greatly reduced in 3D culture, consistent with the reduction of *Mtap* transcription in 3D vs 2D culture conditions (Figs. 7H). We further verified this by quantitative realtime PCR (Fig. 7I) and western blot analysis (Fig. 7J). To determine the mechanism by which *Mtap* was reduced in 3D culture, we examined the chromatin accessibility at the *Mtap* genomic locus. The results showed a drastic effect of culture conditions on *Mtap* chromatin accessibility in which four

open chromatin regions were closed in 3D culture (Fig. 7K). These data suggest that the enhancer activity driving *Mtap* expression was repressed, leading to the downregulation of *Mtap* expression, and consequently, synthetic lethality with partial loss of *Prmt5*.

Discussion

The heterogeneity of an organ or tumor at single cell level is not only determined by genetic and epigenetic factors but also by its surrounding microenvironment such as nutrient and oxygen availability and cell-cell interaction. We do not entirely understand how each cell adapts to its endogenous and exogenous milieus for cellular fitness. While technologies such as scRNA-seq and spatial transcriptomics have helped to understand cellular heterogeneity, it is challenging to dissect each cell's fate in an *in vivo* setting like an organ or a tumor mass. Genome-wide CRISPR screening approach in combination with certain environmental conditions in an *in vitro* setting are valuable for understanding the combined genetic and environmental interactions that determine cell fitness. In this study, we used a MYC-driven murine liver cancer model²⁶, and successfully identified context-specific fitness genes and pathways, as well as commonly shared fitness genes, in monolayer culture under 21% and 1% oxygen tensions and 3D spheroid culture under 21% oxygen tension. Notably, our study revealed that (1) organogenesis pathways such as TGFb-SMAD are enriched in 3D spheroids under positive selection; (2) epigenetic modifier genes encoding BCOR, KMT2D, METTL3 and METTL14 act in different ways in 2D vs. 3D culture; (3) Loss of the VHL-HIF1 pathway is incompatible with cell survival in normoxic 2D conditions, but not hypoxic 2D or normoxic 3D conditions; (4) Distinct requirements for each complex of the electron transport chain exist in normoxia, hypoxia and 3D; (5) Distinct requirements for fatty acid and cholesterol synthesis pathways exist in normoxia and hypoxia and 3D; and (6) Epigenetic reprogramming of *Mtap* in 3D culture leads to context-dependent synthetic lethality to *Prmt5* knockdown. Overall, our studies demonstrated that cancer cells have distinct fitness mechanisms which are dependent on culture conditions. While these findings may not be overtly surprising, our study also revealed nuanced, counterintuitive findings such as each component of the same signaling pathway (i.e., complex of oxidative phosphorylation) exhibiting distinct effects on cell fitness when genetically deleted. Nevertheless, the question of why knockout of each complex of the electron chain reaction gave rise to different fitness outcomes under different cellular context remains to be answered.

Epigenetic modifiers such as KMT2D, BCOR, METTL3 and METTL14 were a limiting factor for uncontrolled cell proliferation in 3D spheroids, which may reveal the mechanism of how they function as tumor suppressors in human cancer. These epigenetic modifiers likely maintain the cellular homeostasis during organogenesis, and when disrupted, tumorigenesis ensues. Surprisingly, knockout of *Kmt2d* and *Mettl3/Mettl14* led to fitness defects in 2D culture under 21% oxygen tension. The DepMAP data also showed that cells cannot survive when *KMT2D* is deleted regardless of its mutation status. We speculate that cell-cell communication in 3D culture more closely represents organogenesis *in vivo*, which needs to be well controlled to prevent aberrant growth. While under a 2D setting, such cell-cell communications do not exist, alternative signaling acts and gives rise to different phenotypes from 3D culture. While the mechanisms accounting for the phenotypic discrepancies in 2D vs 3D conditions for these epigenetic modifiers await elucidation in future studies, our current findings demonstrate that caution should be taken when interpreting the phenotypic screening of these epigenetic modifiers under conventional cell culture conditions.

While HIF pathways were induced by both 1% oxygen and 3D culture, CRISPR screening showed that fewer HIF targets constitute fitness genes in this model system. Instead, hypoxia-inducible genes that were not HIF targets were selectively essential at different oxygen tensions. These results were consistent with previous studies^{12,13}. Nevertheless, we found that *Vhl* and *Hif1a* were essential for cell survival in NEJF10 cells under 21% and 1% oxygen respectively, and this

was not observed in K562 and U2OS cells^{12,13}, suggesting a cell type-specific effect. The previous study found that lipid metabolism and peroxisome genes in K562 cells are essential in hypoxia¹³. However, the peroxisome pathway was not enriched in the MYC-driven NEJF10 cells in either culture condition, suggesting that peroxisome-mediated lipid metabolism might not be essential to MYC-driven cancer. Interestingly, genes responsible for saturated (*Fasn*, *Acaca*) and non-saturated fatty acid synthesis (*Scd2*) or fatty acid catabolism (*Acs14*) exert opposite functions in cell fitness. Considering that both MYC and HIF play critical roles in regulating metabolic gene expression, further dissection of their interaction under different cellular contexts may help us understand the context-specific fitness genes.

Methods and materials

Established cell lines and spheroid culture

NEJF10, NEJF6 and HepG2 (ATCC, HB-8065) cells were maintained in DMEM (Fisher Scientific, Cat#MT10013CM) supplemented with 10% FBS (Gibco, Cat#10437028), and 1% Penicillin-Streptomycin solution (Gibco, Cat#15140122) at 37°C in 5% CO₂ in a humidified incubator. U2OS (ATCC, HTB96) and HCT116 (ATCC, CCL-247) were cultured with McCoy's 5A (Fisher Scientific, 16600082) supplemented with 10% FBS (Gibco, Cat#10437028), and 1% Penicillin-Streptomycin solution (Gibco, Cat#15140122) at 37°C in 5% CO₂ in a humidified incubator. The spheroids formed when NEJF10 or NEJF6 liver cancer cell lines were cultured in low-attachment dish or flask with standard complete DMEM media as mentioned above for 2D cell culture.

Modified cell lines

Generation of NEJF10 cells with PRMT5 shRNA Knockdown

Plasmids of shRNA-ctrl (Vectorbuilder, VB010000-0009mxc), PRMT5-shRNA#1 (Vectorbuilder, VB900070-3948cva), PRMT5-shRNA#2 (Vectorbuilder, VB900070-3942srh), and PRMT5-shRNA#3 (Vectorbuilder, VB900080-7913zan) were purchased from Vector Builder. Plasmids were maxiprepred by using NucleoBond Xtra EF kits (Takara Bio USA, 740424-50) following manufacturer's protocol. shRNA Lentivirus were produced with transient transfection of PEI-pro plasmids complex (6µg of shRNA-ctrl, PRMT5shRNA#1, shRNA#2, shRNA#3, 3 µg of 1-1r, 1 µg RTR, 1 µg of VSVg with 22 ul of PEI pro in 400 µl of DMEM medium) with 5 × 10⁶ HEK293T cells in 10 ml complete medium (DMEM, 10% FBS and 100 U/mL penicillin/streptomycin) in a 10 cm dish. Virus supernatant was collected every 8–12 hours for 3 days, which were passed through a 0.45µm filter and concentrated by ultracentrifuge at 28,500 rpm for 2 hours at 4°C. NEJF10 cell was seeded in six well plate one day before lentivirus transduction. The lentivirus particles were added to NEJF10 cells with polybrene to final concentration of 8 µg/ml. Puromycin (4 µg/ml in complete medium) selection were performed in the next day after virus transduction. After three days selection, NEJF10-shRNA-ctrl, NEJF10-PRMT5-shRNA#1, NEJF10-PRMT5-shRNA#2 and NEJF10-PRMT5-shRNA#3 post selection cells were cultured and expanded to perform 2D crystal violet staining under hypoxia and normoxia culture conditions and 3D organoids proliferation assay with DMEM complete medium without puromycin.

PRMT5 knockout in ABC-Myc genetic mouse model

Albumin-Cre (Alb-Cre) (Strain #003574), R26StopFLMYC (CAG-MYC) (Strain #020458), and PRMT5 flox (Strain # 034414) mice were obtained from the Jackson Laboratory. To generate Prmt5 specific KO in mouse liver, Alb-Cre^{+/+} mice were bred with Prmt5^{flox/flox} mice to get Alb-Cre^{+/wt} ::Prmt5^{flox/wt} mice. CAG-MYC^{+/+} mice were bred with Prmt5^{flox/flox} mice to obtain CAG-MYC^{+/wt} ::Prmt5^{flox/wt} mice. Then, Alb-Cre^{+/wt} ::Prmt5^{flox/wt} mice were bred with CAG-MYC^{+/wt} ::Prmt5^{flox/wt} to generate Alb-Cre^{+/wt} :: CAG-Myc^{+/wt} ::Prmt5^{wt/wt} (ABC-MYC), Alb-Cre^{+/wt} ::

CAG-Myc^{+/wt}::Prmt5^{wt/wt} (ABC-MYC), Alb-Cre^{+/wt}::CAG-Myc^{+/wt}::Prmt5^{flox/wt} (ABC-MYC-Prmt5^{flox/wt}), Alb-Cre^{+/wt}::CAG-Myc^{+/wt}::Prmt5^{flox/flox} (ABC-MYC-Prmt5^{flox/flox}), Alb-Cre^{+/wt}::CAG-Myc^{+/wt}::Prmt5^{flox/wt} (Alb-Cre-Prmt5^{flox/wt}), Alb-Cre^{+/wt}::CAG-Myc^{+/wt}::Prmt5^{flox/flox} (Alb-Cre-Prmt5^{flox/flox}). For genotyping, the genomic DNA was extracted from tail biopsies, and PCR amplification assay was performed using KAPA Mouse Genotyping Kits (Roche Corporate, Cat#KK7352) according to The Jackson Laboratory genotyping PCR conditions for each mice strain. The primers 5'-TGC AAA CAT CAC ATG CAC AC, GAA GCA GAA GCT TAG GAA GAT GG-3' and 5'-TTG GCC CCT TAC CAT AAC TG-3' were used for Alb-Cre genotyping (AlbCre = 390bp and WT = 351bp). The primers 5'-CCA AAG TCG CTC TGA GTT GTT ATC-3', 5'-GAG CGG GAG AAA TGG ATA TG-3', 5'-CCA AGA GGG TCA AGT TGG A-3' and 5'-GCA ATA TGG TGG AAA ATA AC-3' are used for CAG-Myc genotyping (MYC = 550bp and WT = 604bp). The primers 5'-GAT ACC AGG ACC CAC ACT GG-3', and 5'-CTT AGG GGC TTC ATG CAT CT-3' were used for Prmt5 flox genotyping (flox ~ 330 bp and wild type = 239 bp). The genotyping PCR products were resolved in 2% agarose gel (Invitrogen, Cat#16500-500) and imaged with Li-COR D-Digit (Li-COR, 3500). Mice were housed with temperature and 12h light /12h dark cycle controlled under specific-pathogen-free conditions (SPF) at the St Jude Children's Research Hospital mouse facility. All experiments that involved the use of mice were performed in accordance with the guidelines outlined by the St Jude Children's Research Hospital Institutional Animal Care and Use Committee (IACUC).

Method details

Seahorse real-time ATP rate assay

The ATP production rate assay was determined using the Seahorse XF Real-Time ATP Rate Assay Kit (Agilent, 103592-100) and the Seahorse XF Pro Analyzer (Agilent). Briefly, cells were seeded into the Seahorse XF Pro M cell culture microplate (Agilent, 103774-100) at different cell densities (10000, 20000, and 4000 cells per well). At the same time, the XF Pro Sensor Cartridge was hydrated using 200 µl of XF calibrant (Agilent, 100840-000) in a 37°C CO₂-free incubator overnight. The next day, the cells were washed and incubated with XF DMEM medium (Agilent, 103575-100) supplemented with 10 mM glucose (Agilent, 103577-100), 1 mM pyruvate (Agilent, 103578-100), and 2 mM L-glutamine (Agilent, 103579-100). OCR and ECAR were measured in the Agilent's Seahorse XF Pro Extracellular Flux Analyzer by subsequent sequential injections of two compounds that affect the cellular bioenergetic processes, as follows: 20 µl of oligomycin (10 µM) in port A and 22 µl of rotenone/antimycin A (5 µM) in port B. Data was processed with Seahorse Analytics.

RNA-seq analysis for differential gene expression (DGE)

NEJF10 cells were cultured with DMEM complete medium in 6 well plate (Falcon, 353046) for 2D and in ultralow attachment 6 well plate (Corning, 3471) for 3D under 21% and 1% oxygen conditions for 48 hours. Total RNA extraction from cells was performed using the RNeasy Mini Kit (74106, Qiagen) according to the manufacturer's instructions.

Total stranded RNA sequencing data were processed by the internal AutoMapper pipeline. Briefly the raw reads were first trimmed (Trim-Galore version 0.60), mapped to mouse genome assembly (mm10) (STAR v2.7) and then the gene level values were quantified (RSEM v1.31) based on GENCODE annotation (M22). Low count genes were removed from analysis using a CPM cutoff corresponding to a count of 10 reads and only confidently annotated (level 1 and 2 gene annotation) and protein-coding genes are used for differential expression analysis. Normalization factors were generated using the TMM method, counts were normalized using voom and normalized counts were analyzed using the lmFit and eBayes functions (R limma package version 3.6.3). The significantly up- and down-regulated genes were defined by at least 2-fold changes and adjusted p-value < 0.05. Then Gene Set Enrichment Analysis (GSEA) was conducted using gene-level log2 fold changes from differential expression results against gene sets in the Molecular Signatures Database (MSigDB 6.2) (gsea2 version 2.2.3).

RNA-seq analysis for alternative splicing analysis

After mapping RNA-seq data, rMATS v4.1.0 was used for RNA alternative splicing analysis by using the mapped BAM files as input. Specifically, five different kinds of alternative splicing events were identified, i.e., skipped exon (SE), alternative 5'-splicing site (A5SS), alternative 3'-splicing site (A3SS), mutually exclusive exon (MXE) and intron retention (RI). To keep consistent, the same GTF annotation reference file for mapping was used for rMATS. For stranded RNA-seq data, the argument “--libType fr-firststrand” was applied. To process reads with variable lengths, the argument “--variable-read-length” was also used for rMATS. To select statistically significantly differential splicing events, the following thresholds were used: FDR < 0.05 and the absolute value of IncLevelDifference > 0.1. For visualization, the IGV Genome Browser was used to show the sashimi plots of splicing events.

CRISPR screening and data analysis

The Mouse CRISPR Knockout Pooled Library (Brie, lentiCRISPRv2) was obtained from Addgene (Addgene#73632), which includes 1000 control gRNAs and 78,637 unique sgRNA sequences targeting 19,674 human genes (4 sgRNAs per gene, and 1000 non-targeting controls). The plasmid library was amplified and validated in the Center for Advanced Genome Engineering at St. Jude Children's Research Hospital as described in the Broad GPP protocol (<https://portals.broadinstitute.org/gpp/public/resources/protocols>) except Endura™ DUOs (Lucigen) electrocompetent cells were used for the transformation step. The workflow of this whole genome genetic screen is illustrated in **Fig. 2A**. The NEJF10 cells were transduced with mouse CRISPR Knockout pooled library (Brie) at a low MOI (~ 0.3) to ensure effective barcoding of individual cells. Cells were replenished with fresh DMEM medium containing 2 µg/mL puromycin (Millipore Sigma) for 36 h. After puromycin selection, cells were washed to eliminate dead cell debris and maintained in complete DMEM medium.

In the process of 2D screening, to maintain adequate representation of gRNAs, during the CRISPR screening, a total of 8 million cells were re-plated during each time point, ensuring a consistent 100x coverage of the Brie library. The 2D hypoxia screening took place within a hypoxia chamber, and all cell culture activities were conducted under the controlled environment of this chamber (Whitley H35 HEPA Hypoxystation). In the context of Brie-library screening involving NEJF-10 cells within a 3D culture framework, 0.5 million cells were introduced into each well of a 96-well plate (non-adherent), totaling 144 wells. A centrifugation step at 1000 rpm for 5 minutes was subsequently undertaken. Each individual spheroid maintained an approximate coverage of ~ 6.2x for each gRNA, and when combined, the total gRNA coverage across all spheroids reached 900x. On the following day, the formed spheroids were carefully transferred to a non-adherent T75 flask, ensuring their spherical structure was preserved. The spheroids were cultured within a regular CO₂ incubator, utilizing a shaker operating at 80 rpm. To maintain their growth, the cell culture medium was replenished every alternate day. At each time point 32x10⁶ cells were collected for genomic DNA extraction to ensure over 400x coverage.

The total genomic DNA was extracted using a DNeasy Blood & Tissue Kit (Qiagen) and quantified with a Nanodrop instrument. The sgRNA sequences were amplified using PCR method using NEB Q5 polymerase (New England Biolabs). PCR products were purified by AMPure XP SPRI beads (Beckman Coulter) and quantified by a Qubit dsDNA HS assay (Thermo Fisher Scientific). A total of 16 million reads were sequenced using an Illumina HiSeq sequencer, and the sequencing data were analyzed using MAGeCK-VISPR software⁵⁸. NGS sequencing was performed in the Hartwell Center Genome Sequencing Facility at St. Jude Children's Research Hospital. Single-end, 100-cycle sequencing was performed on a NovaSeq 6000 (Illumina). Validation to check gRNA presence and representation was performed using `calc_auc_v1.1.py` (<https://github.com/mhegde/>) and `count_spacers.py`. Network analysis performed using STRING program (<https://string-db.org/>).

Lentivirus preparation and transduction for VHL knockout in HepG2 cells

The lentiviral-based knockout plasmids targeting the VHL gene (VHL-CRISPR guide RNA1 pLentiCRISPR v2 and VHL-CRISPR guide RNA2 pLentiCRISPR v2, Cat#SC1678) were procured from GenScript. The control plasmid, pLentiCRISPR v2 (Cat#52961), was sourced from Addgene. Plasmids were maxiprep by using NucleoBond Xtra EF kits (Takara Bio USA, 740424-50) following manufacturer's protocol. Lentivirus was produced by transient transfection of PEI-pro DNA complex (LentiV2, VHL-gRNA-1 and VHL-gRNA-2 plasmids, 1-1r, RTR, VSVg with PEI pro in DMEM medium) with HEK293T cells in 12 ml DMEM complete medium in a 15 cm dish. Virus supernatant was collected every 8–12 hours for 3 days, which were passed through a 0.45 μ m filter and concentrated by ultracentrifuge at 28,500 rpm for 2 hours at 4°C. The VHL-gRNA virus particles were added to HepG2 cells which seeded previous day, following add polybrene to final concentration of 8 μ g/ml. Puromycin (1 μ g/ml in complete medium) selection were performed in the next day after virus transduction. At the end of the 72 hours, all HepG2 parental control cells were dead. HepG2-lentiV2, HepG2-VHL-KO1 and HepG2-VHL-KO2 cells post selection were cultured and expanded with DMEM complete medium. For 2D and 3D cell proliferation assay, HepG2-lentiV2, HepG2-VHL-KO1 and HepG2-VHL-KO2 cells were cultured with 96 well clear flat bottom (Corning, 353072) for 2D and 96-well clear round bottom ultra-low attachment plate (Corning, 7007) for 3D in the IncuCyte. High resolution bright field images were captured every 6–8 hours. Analysis modules use edge finding algorithm to quantity 2D and 3D growth and size over time to measure cell proliferation. The raw data of 2D cell proliferation were exported to and plotted with GraphPad Prism 9.5.1. For the 3D cell proliferation, the raw data was exported to excel and normalized to 24 hours 3D size, then plotted graph with GraphPad Prism 9.5.1.

Online bioinformatics tools and programs

Pathway analysis was performed by using Enrichr program (<https://maayanlab.cloud/Enrichr/>). Proteinprotein interaction network was analyzed by using STRING program (<https://string-db.org/>). Correlation of knockout effects of two genes, knockout of one gene vs gene expression, gene knockout effect vs drug effect, gene expression vs drug effect was analyzed by DepMap data (<https://depmap.org/portal/>), then data were downloaded and presented by using PRISM program.

Histology and immunohistochemistry

Liver tumors were fixed in 10% neutral buffered formalin, embedded in paraffin, sectioned at 4 μ m, mounted on positive charged glass slides (Superfrost Plus; 12-550-15, Thermo Fisher Scientific, Waltham, MA) that were dried at 60°C for 20 minutes, and stained with hematoxylin and eosin (HE). The following immunohistochemistry protocol was used for the detection of anti-Galectin-3 (Mac-2) (M3/38, ACL8942AP, Accurate Chemical and Scientific Corporation), 1:1000, 32' incubation, heat-induced epitope retrieval with cell conditioning media 1 (Ventana Medical Systems, Tucson, AZ), 32 minutes; Visualization with DISCOVERY OmniMap anti-Rt HRP (760-4311; Ventana Medical Systems), DISCOVERY ChromoMap DAB kit (760 - 159; Ventana Medical Systems).

Western blot

For western blotting, samples were mixed with equal volume of 2 X sample buffer (1M TRIS/HCl, 10% SDS, 0.1% bromophenol-blue, 10% β - mercaptoethanol, 10% glycerol), sonicated for 30 seconds on 30% AMPL (Sonics & Materials Inc, VCX 130PB) and heated for 15 minutes at 95°C. Proteins were resolved on SDS-polyacrylamide gels (Bio-Rad, Cat#4568083) and transferred onto PVDF membrane (Bio-Rad, Cat#170-4272) with Trans-blot Turbo transfer system (Bio-Rad, Cat#1704150). After being incubated with primary antibodies overnight at -20, horseradish peroxidase-(HRP) conjugated secondary antibody (Thermo Fisher, A245373, A245182) at 1: 5000 was used for 1 hour incubation. The signals were detected by chemiluminescence (ECL, Thermo Fisher). Images were taken using Li-COR Odyssey FC (LiCOR, Cat#2800). Antibodies including β -

Actin (Sigma, A1978, 1:5,000), HIF1 α (CAYMAN, 10006421, 1:200), HIF2 α (Novus, NB100-122, 1:500), VHL (Cell Signaling Technology, 68547S, 1:1000), PRMT5 (ABclonal, A1520, 1:1000), HSP90 (Santa Cruz, sc13119, 1:1000) and MTAP (Cell Signaling Technology, 4158S, 1:1000) were used for western blot.

Crystal Violet Staining

NEJF10-shRNA-ctrl, NEJF10-PRMT5-shRNA#1, NEJF10-PRMT5-shRNA#2 and NEJF10-PRMT5-shRNA#3 was seeded in 6 well plate (Falcon, 353046) and cultured for 6 days. After removing medium, cells were washed with Dulbecco's phosphate buffered saline without calcium or magnesium (DPBS, Lonza) and treated with 4% Formaldehyde in PBS (PFA) for 20 minutes. Once PFA was removed, cells were stained with 0.1% crystal violet stain for 1 hour, followed by water rinsing three times. The crystal violet staining plates were imaged after the plates were completely dry.

Organoids proliferation assay of PRMT5 knockdown in NEJF10 cell organoids

NEJF10-shRNA-ctrl, NEJF10-PRMT5-shRNA#1, NEJF10-PRMT5-shRNA#2 and NEJF10-PRMT5-shRNA#3 3000 cells per well were cultured with 96-well clear round bottom ultra-low attachment plate (Corning, 7007) in the IncuCyte. IncuCyte scanning was used to live imaging and measure the proliferation and dynamics of PRMT5 knockdown in NEJF10 cell organoids. Replicates were 6–8 per group. High resolution bright field images were captured every 6–8 hours without human interaction. The organoids proliferation was analyzed and quantified using IncuCyte software. Analysis modules use edge finding algorithm to quantify organoid growth and size over time to measure organoid proliferation. Real-time analysis is performed over 9 days. The raw data were exported and plotted with GraphPad Prism 9.5.1.

RT-PCR for MTAP

Total RNA from NEJF10 cells cultured under 2D normoxia, hypoxia and 3D normoxia conditions were performed using the RNeasy Mini Kit (Qiagen) according to the manufacturer's instructions. cDNA was prepared from extracted RNA using Invitrogen™ SuperScript™ IV Reverse Transcriptase (ThermoFisher Scientific, 18091050) and detected by fast SYBR Green (Applied Biosystems, 4368708) on Applied Biosystems 7500 Fast Real-time PCR System. Two sets of MTAP primers (set1: Forward 5'-ACGGCGGTGAAGATTGGAATA-3' and Reverse 5'-ATGGCTTGCCGAATGGAGTAT-3' and Set2: Forward 5'-AAGCCATCCGATGCCTTAATTT-3' and Reverse 5'-TTGCTGGTAGTTGACTTTTGAA-3') were used for RT-PCR and 18s primers (Forward 5'-GCTTAATTTGACTCAACACGGGA-3' and Reverse 5'-AGCTATCAATCTGTCAATCCTGTC-3') were performed as internal controls. qPCR signal for each gene was normalized to those of 18s using the Δ CT method. Results were represented as fold expression relative to WT with the standard error for 3–4 biological replicates.

ATAC-seq

NEJF10 cells were cultured in 21% and 1% oxygen in monolayer, or 21% oxygen in 3D spheroid. Fresh cultured NEJF10 cells (100,000 per sample) under different culture conditions were harvested and washed with 150 μ l cold Dulbecco's Phosphate-Buffered Saline (DPBS) containing protease inhibitor (PI). Nuclei were collected by centrifuging at 500 g for 10 minutes at 4°C after cell pellets were resuspended in lysis buffer (10 mM Tris-Cl pH 7.4, 10 mM NaCl, and 3 mM MgCl₂ containing 0.1% NP-40 and PI). Nuclei were incubated with Tn5 transposon enzyme in transposase reaction mix buffer (Illumina) for 30 min at 37°C. DNAs were purified from transposition sample by using Min-Elute PCR purification kit (Qiagen, Valencia, CA) and measured by Qubit. Polymerase chain reaction (PCR) was performed to amplify with High-Fidelity 2X PCR Master Mix [72°C/5mins + 98°C/30 s + 12 $\times$ (98°C/10 s + 63°C/30 s + 72°C/60 s) + 72°C/5 min]. The libraries were purified using Min-Elute PCR purification kit (Qiagen, Valencia, CA).

ATAC-seq libraries followed by pair-end sequencing on HiSeq4000 (Illumina) in the Hartwell Center at St Jude Children's Research Hospital. The ATAC-seq raw reads were aligned to the mouse reference genome (mm10) using BWA⁵⁹ and then marked duplicated reads with Picard (version 1.65), with only high-quality reads kept by samtools (version 1.3.1, parameter “-q 1 -F 1024”)⁶⁰. Reads mapping to mitochondrial DNA were excluded from the analysis. All mapped reads were offset by + 4 bp for the + strand and -5 bp for the - strand⁶¹. Peaks were called for each sample using MACS2⁶² with parameters “-q 0.01 -nomodel -extsize 200 -shift 100”. Peaks were merged for the same cell types using BEDtools⁶³. Peak annotation was performed using HOMER⁶⁴. All sequencing tracks were viewed using the Integrated Genomic Viewer (IGV 2.3.82)⁶⁵.

Statistical analysis

All quantitative data are presented as mean $\pm$ SD. Unpaired Student's t test was performed for comparison of two groups. Kaplan-Meier method was used to estimate the survival rate. P-values across multiple time points were adjusted for multiple comparison using the Holm-Sidak method. $p < 0.05$ was considered as statistically significant. All the statistical analyses, except where otherwise noted, were performed using GraphPad Prism (v9).

Data and Materials Availability

RNA-seq and ATAC-seq have been deposited in GEO database with accession number: GSE240980, GSE262074. All study data are included in the article and Supplemental Figures and Tables. Cell lines and genetically engineered cell lines will be freely distributed upon request.

Figure supplements

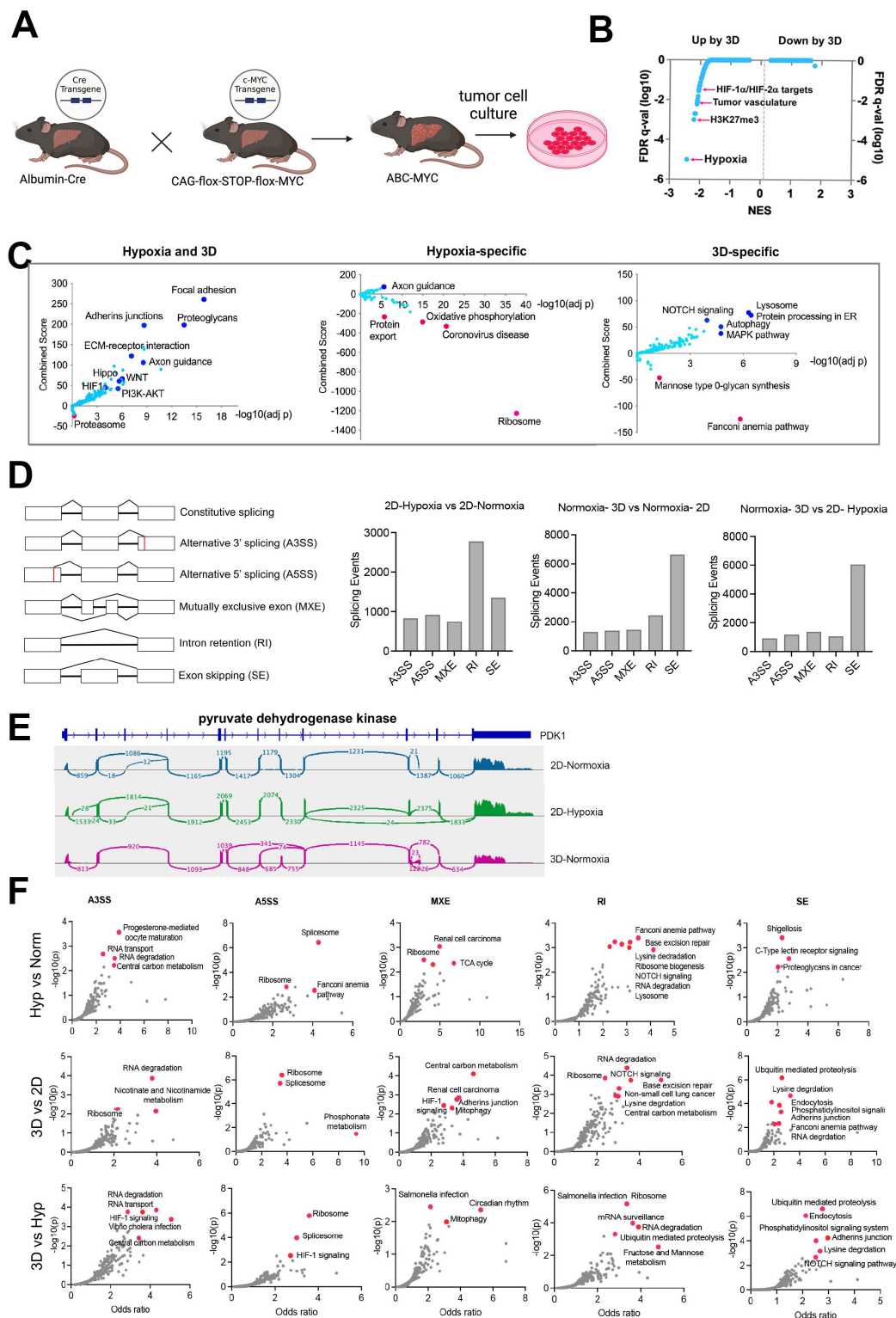


Figure 1- figure supplement 1.

Cells undergo different transcriptional programing in hypoxia, normoxia, and 3D Culture.

(A) Cartoon indicate the generation of ABC-Myc driven hepatoblastoma model for in vitro tumor cell culture. (B) GSEA results for pathway enrichment in 3D vs 2D normoxia. Each dot represents one geneset. NES =normalized enrichment score. FDR =false discovery rate. (C) The volcano plots, generated using Enrichr, show significant KEGG pathways shared by cells in 3D and hypoxia (left), specific to hypoxic (middle) and 3D conditions (right). The X-axis represents the significance of expression change for a gene in $-\log_{10}$ (adjusted P-value). The Y-axis represents the combined score for enrichment of a given pathway with positive values indicating enrichment and negative values indicating a decrease in the pathway. (D) Five major alternative splicing events are illustrated with the boxes representing exons and the connecting lines representing introns. When comparing 2D hypoxia to 2D normoxia, there is high intron retention as indicated by the number of splicing events is over 2000. Normoxic 3D compared to 2D also shows an increase in intron retention along with upregulation of exon skipping. The graph representing 3D Normoxia and 2D Hypoxic shows both conditions upregulate exon skipping; however, increased intron retention is not seen. (E) The splicing events of pyruvate dehydrogenase kinase, a regulator of pyruvate dehydrogenase and known oncogenic driver, is shown in three conditions of interest: 2D normoxia, 2D hypoxia, and 3D normoxia. (F) Pathways for genes undergoing alternative splicing in each category. The Y-axis represents the significance of expression change for a gene in $-\log_{10}$ (adjusted P-value). The X-axis represents the Odds ratio.

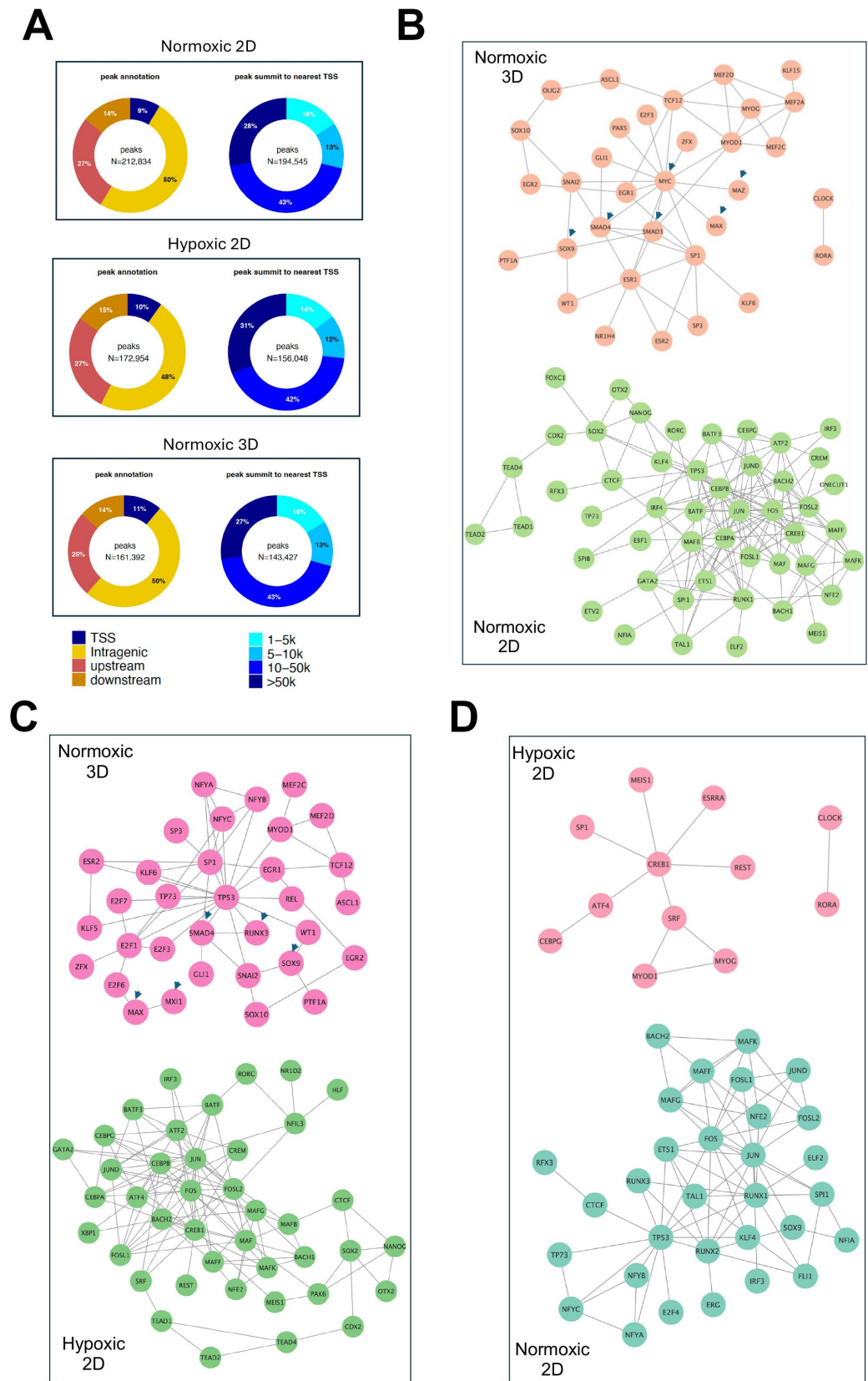


Figure 1- figure supplement 2.

Epigenetic reprogramming under hypoxic 2D and normoxic 3D conditions.

(A) The percentage of ATAC-seq peaks at different genomic regions (transcription start site, intragenic, upstream of 5' and downstream of 3')(left panel), and the percentage of peaks with distances from the transcription start site (TSS). (B-D) Pairwise comparison of the protein-protein interaction networks (STRING confidence threshold = 0.7) formed by transcription factors that are predicted to be highly active under normoxic 2D, hypoxic 2D and normoxic 3D, respectively.

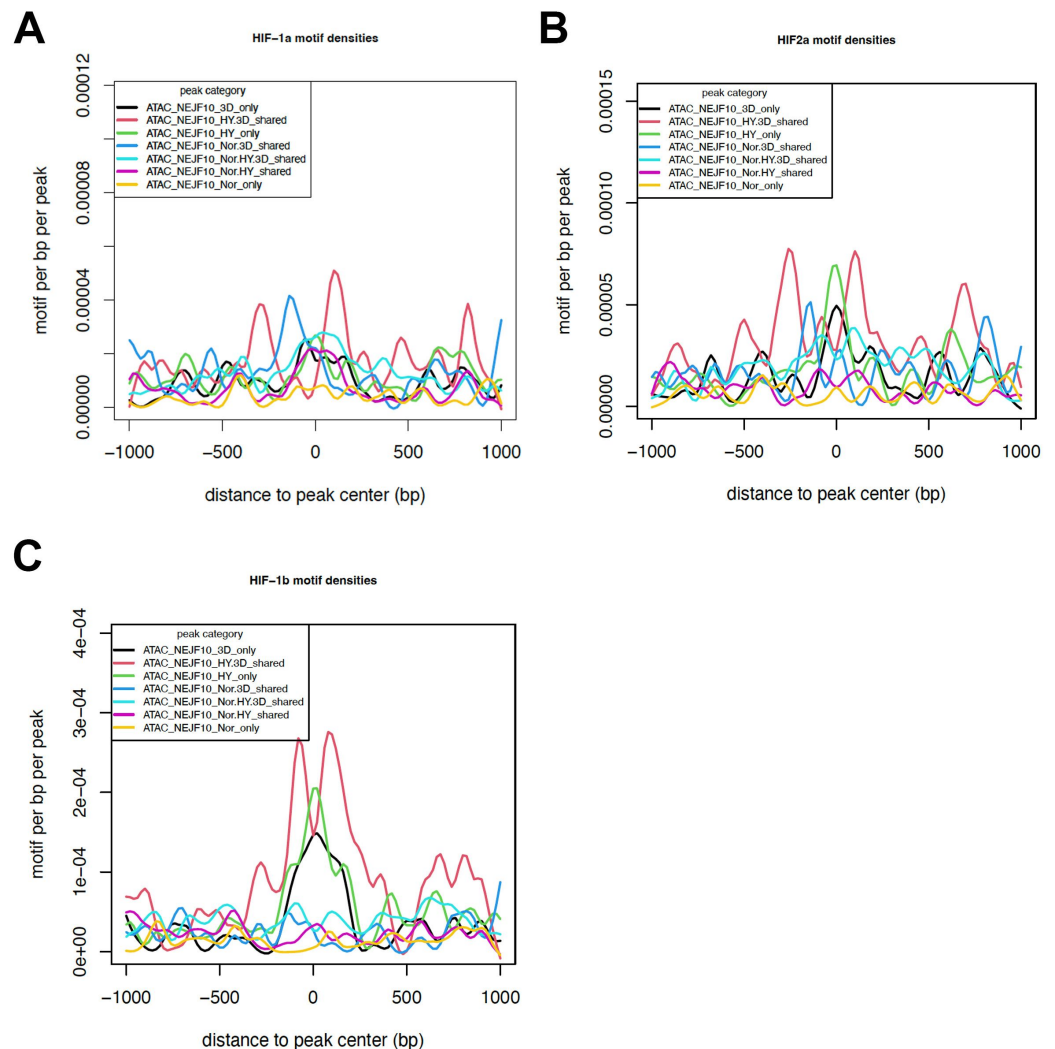


Figure 1- figure supplement 3.

HIF motif densities surrounding the ATAC-seq peaks.

HIF-1a (A), HIF-2a (B) and HIF-1b (C) motif densities around ATAC-seq open chromatin regions (± 1000 bp) by categories. Motif density was determined by HOMER (Hypergeometric Optimization of Motif EnRichment) program and normalized to that in a background sequence of equal length. X-axis indicates the distance to peak center (bp).

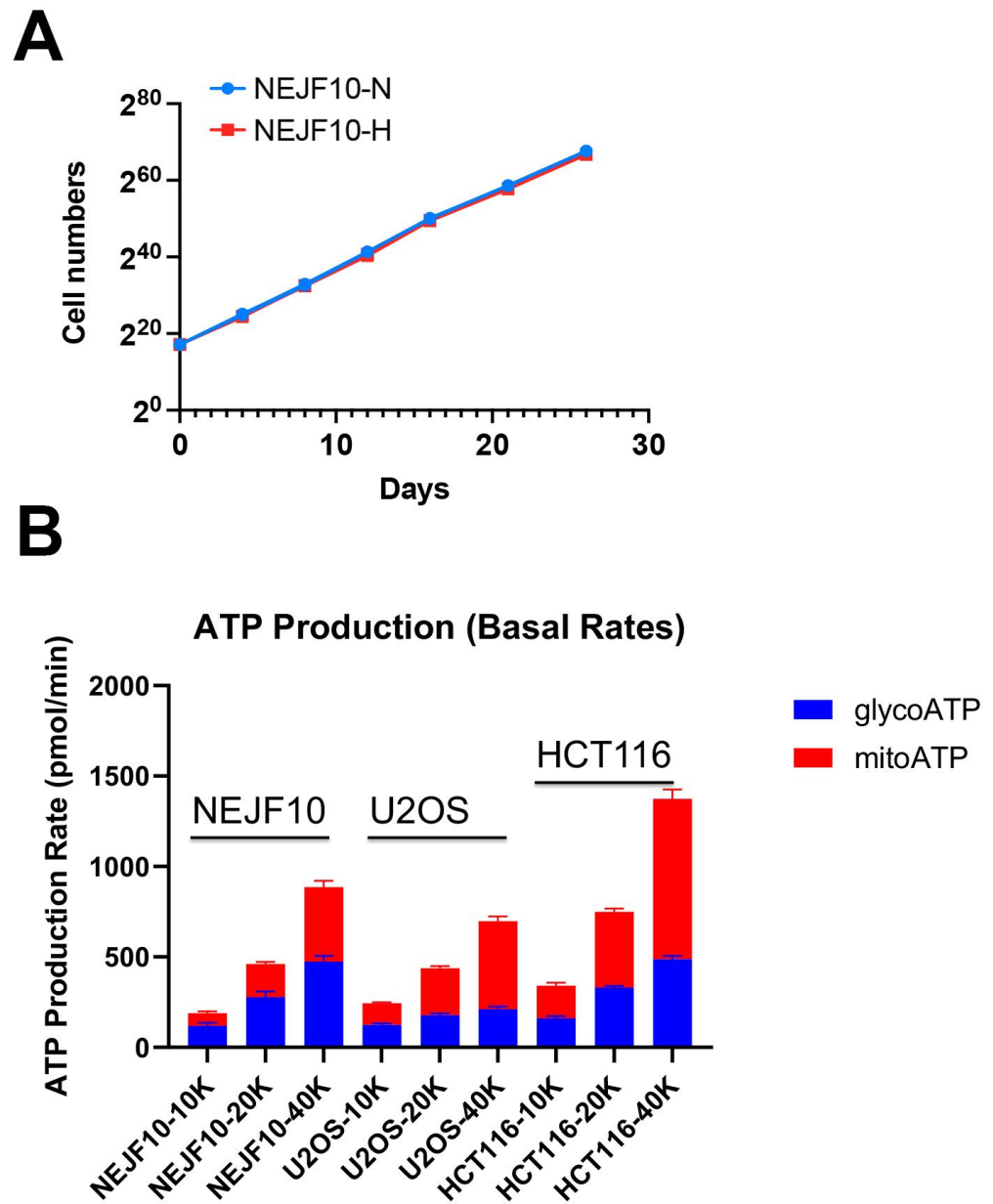


Figure 2- figure supplement 1.

NEJF10 cells tolerate chronic hypoxia and more rely on ATP from glycolysis.

(A) Cell counts over time for NEJF10 cells cultured in 2D under 1% oxygen (hypoxia) and normoxia. (B) Seahorse measurement of ATP sources of NEJ10 cells seeded with different cell numbers (10k, 20k, 40k), U2OS cells and HCT116 cells cultured in 2D normoxia.

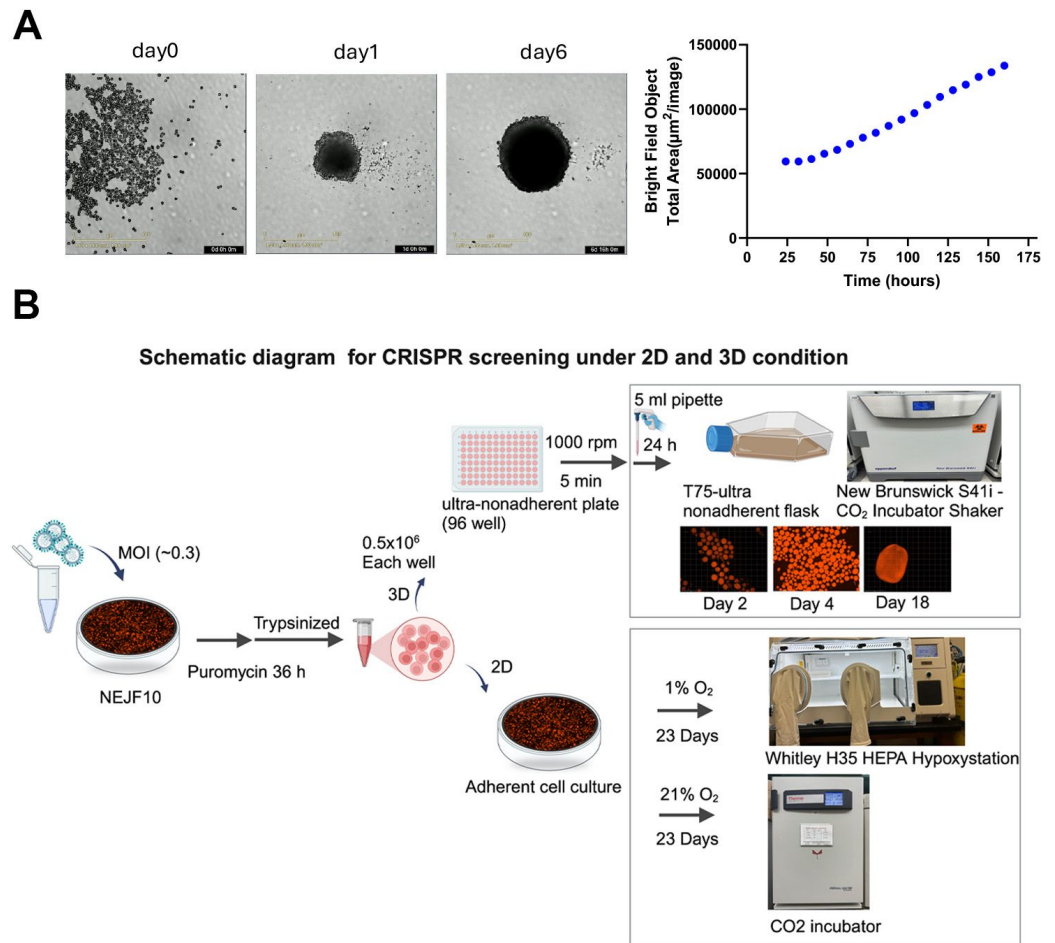


Figure 2- figure supplement 2.

NEJF10 spheroid growth and CRISPR procedure.

(A) Representative images at different time points for 3D growth of NEJF10 seeded with 3000 cells per well in 96 well plate (left). The 3D growth curve monitored by InCuCyte.(right). (B) The schematic showing the CRISPR procedure for NEJF10 cells cultured under different conditions in T75 flask.

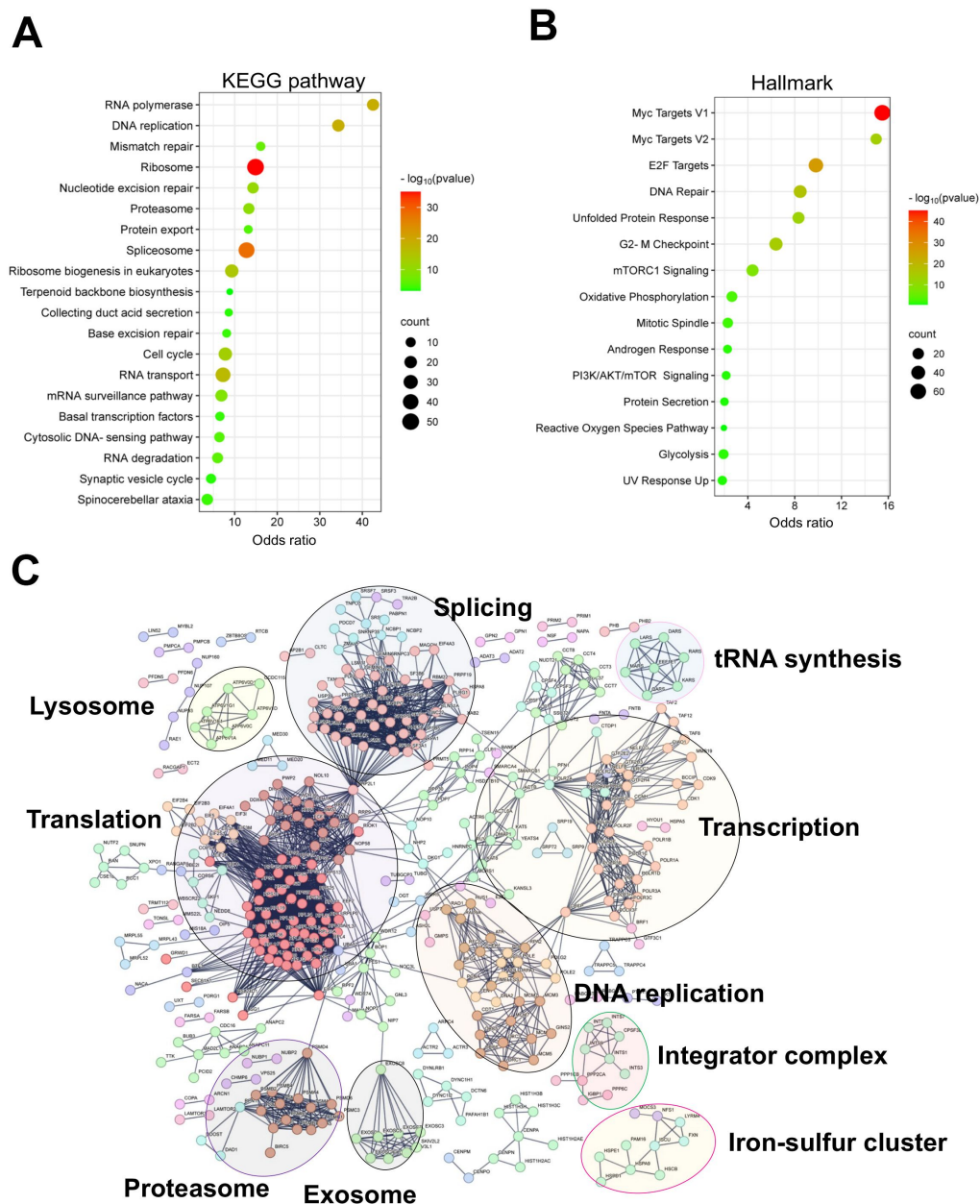


Figure 2- figure supplement 3.

Shared essential genes under normoxia 3D, normoxia 2D and hypoxia 2D.

(A) Bubble blot showing KEGG pathway enrichment for the shared essential genes under normoxia 3D, normoxia 2D and hypoxia 2D. (B) Bubble blot showing hallmark pathway enrichment for the shared essential genes under normoxia 3D, normoxia 2D and hypoxia 2D. (C) STRING interaction network analysis for the shared essential genes under normoxia 3D, normoxia 2D and hypoxia 2D.

Figure 2- figure supplement 4.

Shared essential genes under normoxia 3D and hypoxia 2D.

(A) Bubble blot showing hallmark (top) KEGG (bottom) pathway enrichment for the shared essential genes under normoxia 3D and hypoxia 2D. (B) STRING interaction network analysis for the shared essential genes under normoxia 3D and hypoxia 2D.

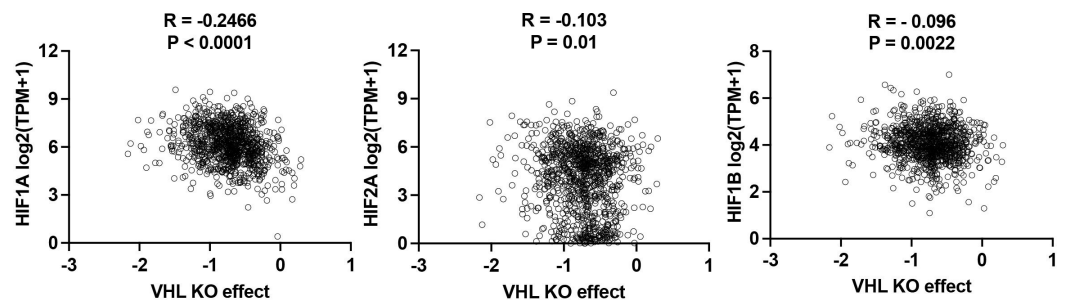


Figure 3- figure supplement 1.

Spearman correlation analysis of VHL knockout effect vs HIF.

Spearman correlation analysis of *VHL* knockout effect vs expression of *HIF-1A*, *HIF-2A* and *HIF-1B* in 2D normoxia. Data were extracted from DepMap (depmap.org).

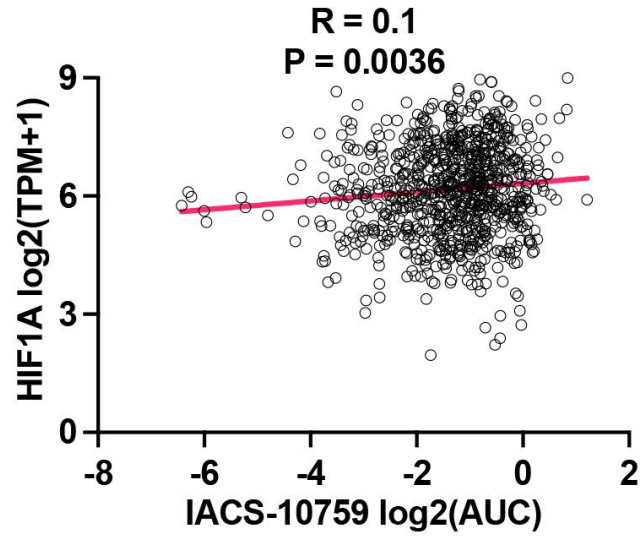


Figure 4- figure supplement 1.

Spearman correlation analysis of IACS-10759 effect vs HIF1A expression.

Spearman correlation analysis of IACS-10759 effect vs expression of *HIF-1A* in 2D normoxia. Data were extracted from DepMap (depmap.org).

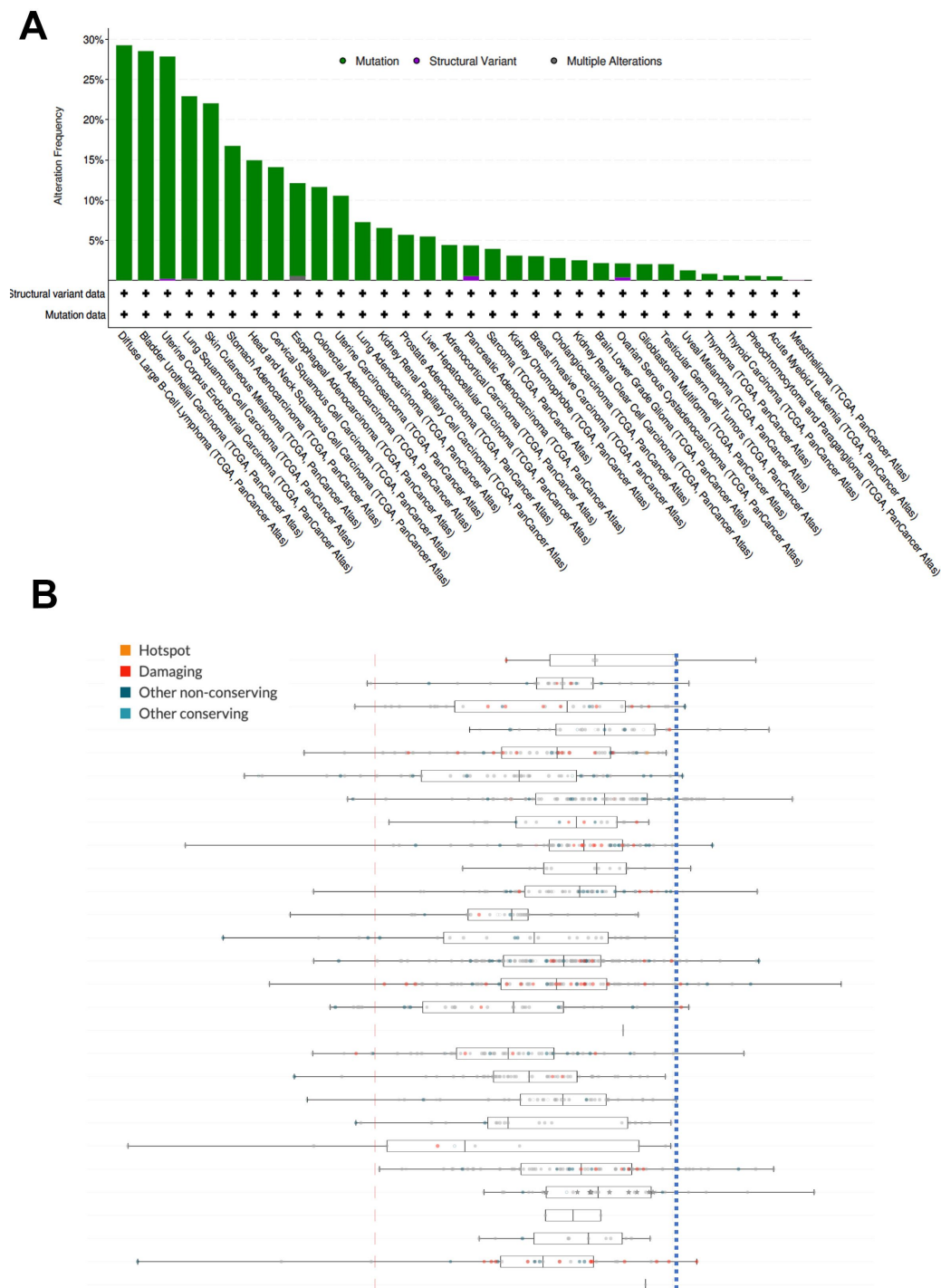


Figure 5- figure supplement 1.

Mutations in *KMT2D* in human cancers and its knockout effect in human cancer cell lines.

(A) Genetic mutations of *KMT2D* across different types of human cancer. Data were extracted from cBioportal (<https://www.cbioportal.org>). (B) Knockout effect of *KMT2D* across different types of human cancer cell lines in 2D normoxia. Data were extracted from DepMap. Red color indicates mutations. The blue dotted line indicates CRISPR effect as "0".

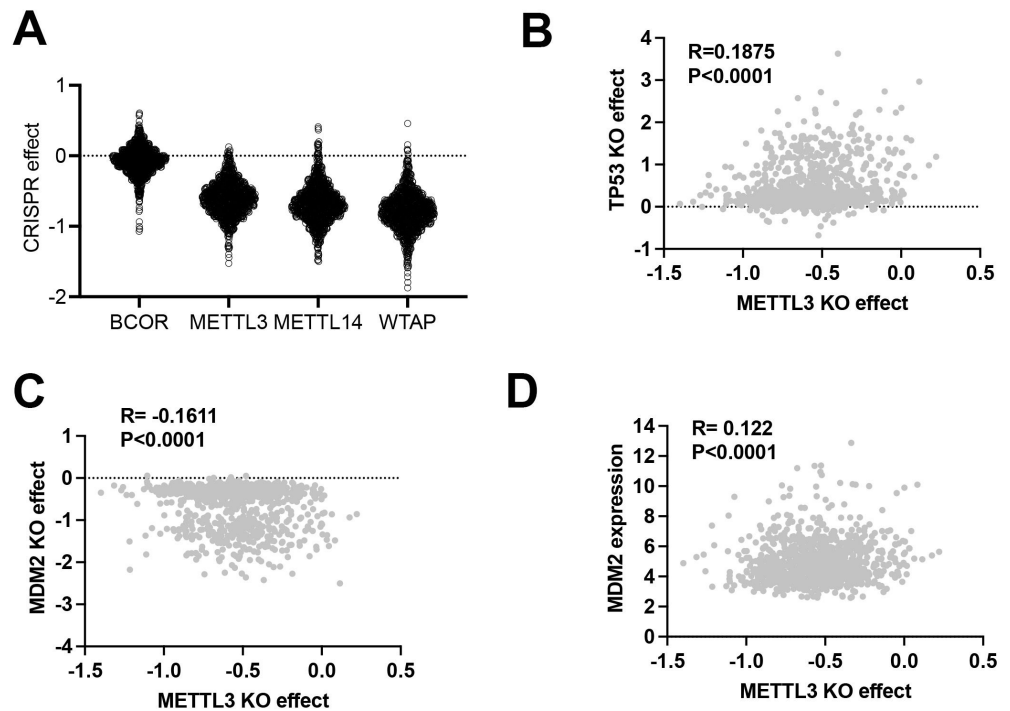


Figure 5- figure supplement 2.

Gene knockout effect in human cancer cell lines under 2D normoxia.

(A) CRISPR knockout effect of *BCOR*, *METTL3*, *METTL14* and *WTAP* across different types of human cancer cell lines in 2D normoxia. Data were extracted from DepMap. (B) Spearman correlation of knockout effect for *TP53* vs *METTL3* across different types of human cancer cell lines in 2D normoxia. Data were extracted from DepMap. (C) Spearman correlation of knockout effect for *MDM2* vs *METTL3* across different types of human cancer cell lines in 2D normoxia. Data were extracted from DepMap. (D) Spearman correlation of *METTL3* knockout effect vs *MDM2* expression across different types of human cancer cell lines in 2D normoxia. Data were extracted from DepMap.



Figure 6- figure supplement 1.

Pathways enrichment for fitness genes whose expression is altered in hypoxia and 3D.

(A) Bubble blot showing pathway enrichment for the essential genes under hypoxia 2D whose expression is also altered in hypoxia 2D. However, this analysis does not exclude the genes that may also be altered under 3D culture.

(B) Bubble blot showing pathway enrichment for the positive selection genes under hypoxia 2D whose expression is also altered in hypoxia 2D. However, this analysis does not exclude the genes that may also be altered under 3D

culture. (C) Bubble blot showing pathway enrichment for the essential genes under 3D whose expression is also altered in 3D. However, this analysis does not exclude the genes that may also be altered under hypoxia 2D culture. (D) Bubble blot showing pathway enrichment for the positive selection genes under 3D whose expression is also altered in 3D. However, this analysis does not exclude the genes that may also be altered under hypoxia 2D culture.

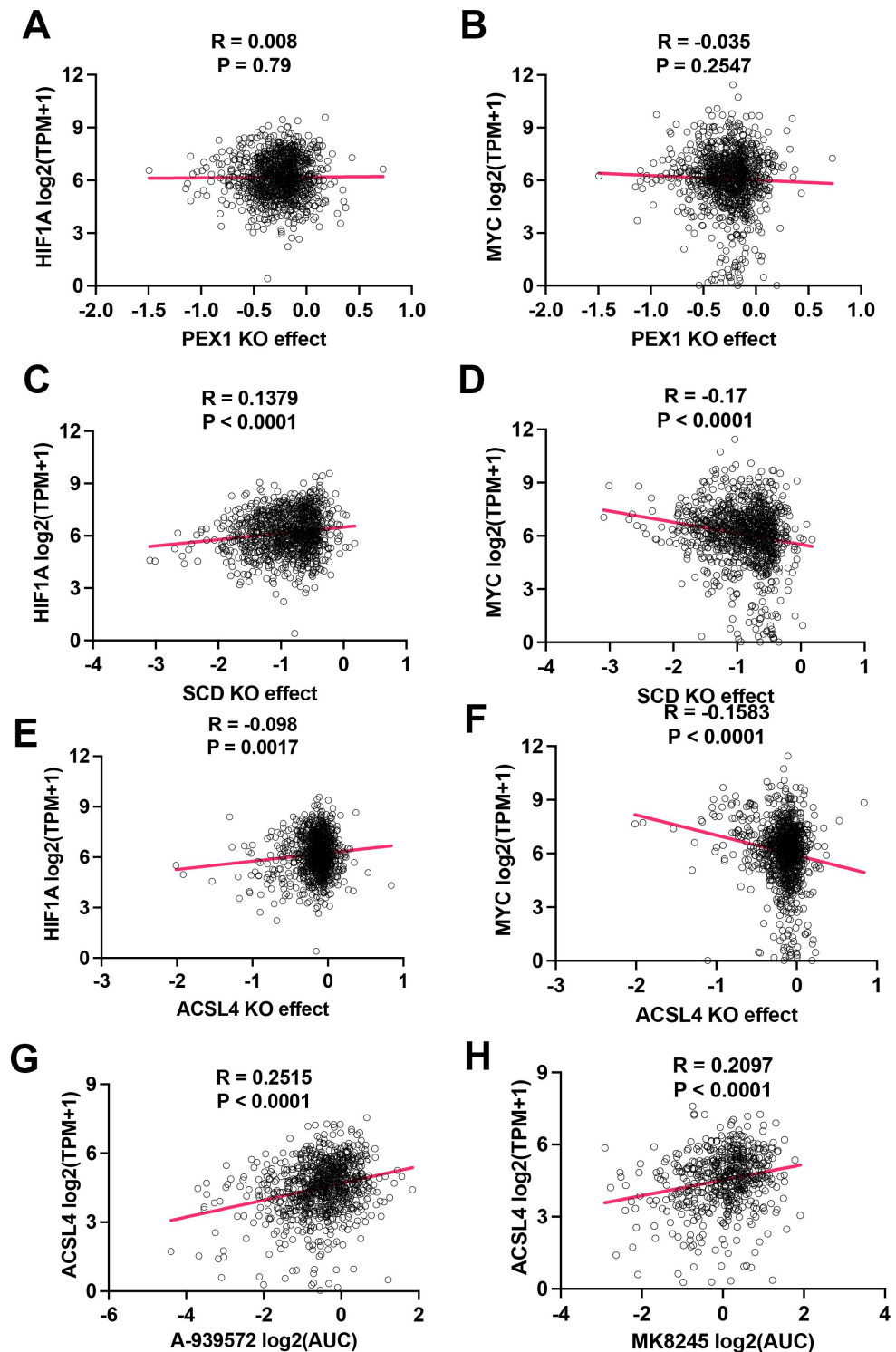


Figure 6- figure supplement 2.

Inhibition effect of lipid metabolic genes of human homologs.

(A) Spearman correlation analysis of *PEX1* knockout effect vs expression of *HIF-1A* in 2D normoxia. Data were extracted from DepMap (depmap.org). (B) Spearman correlation analysis of *PEX1* knockout effect vs expression of *MYC* in 2D normoxia. Data were extracted from DepMap (depmap.org). (C) Spearman correlation analysis of *SDC* knockout effect vs expression of *HIF-1A* in 2D normoxia. Data were extracted from DepMap (depmap.org). (D) Spearman correlation analysis of *SDC* knockout effect vs expression of *MYC* in 2D normoxia. Data were extracted from DepMap (depmap.org). (E) Spearman correlation analysis of *ACSL4* knockout effect vs expression of *HIF-1A* in 2D normoxia. Data were extracted from DepMap (depmap.org). (F) Spearman correlation analysis of *ACSL4* knockout effect vs expression of *MYC* in 2D normoxia. Data were extracted from DepMap (depmap.org). (G) Spearman correlation analysis of A-939572 effect vs expression of *ACSL4* in 2D normoxia. Data were extracted from DepMap (depmap.org). (H) Spearman correlation analysis of MK8245 effect vs expression of *ACSL4* in 2D normoxia. Data were extracted from DepMap (depmap.org).

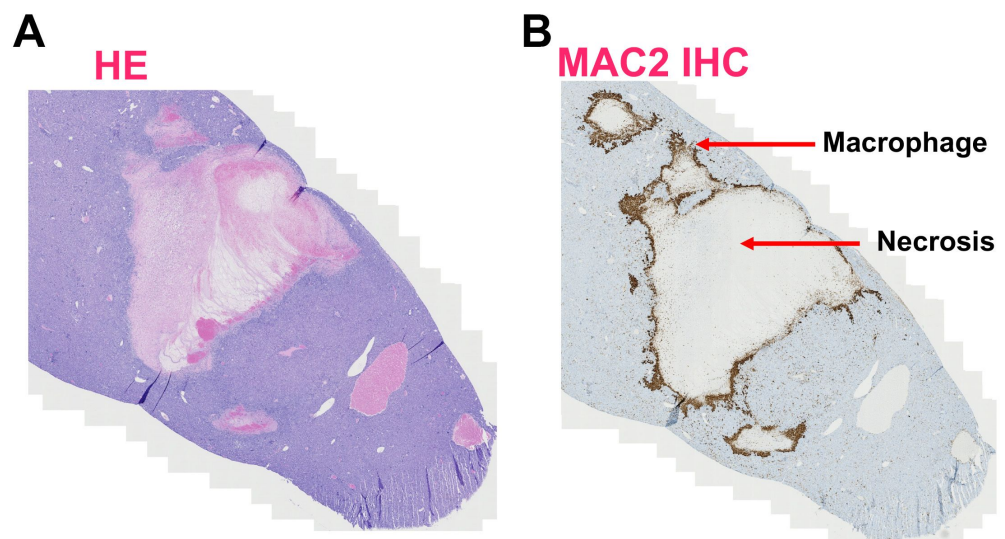


Figure 7- figure supplement 1.

Necrosis and inflammation of liver tumor with *Prmt5* knockout.

(A) H&E staining of ABC-MYC liver tumors with *Prmt5* knockout. (B) MAC2 immunohistochemistry staining of ABC-MYC liver tumors with *Prmt5* knockout.

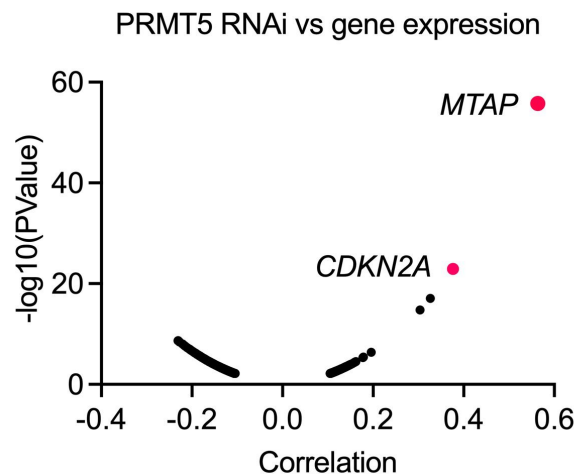


Figure 7- figure supplement 2.

PRMT5 knockdown effect vs MTAP expression.

Spearman correlation of knockdown effect for *PTMR5* vs *MTAP* expression across different types of human cancer cell lines in 2D normoxia. Data were extracted from DepMap.

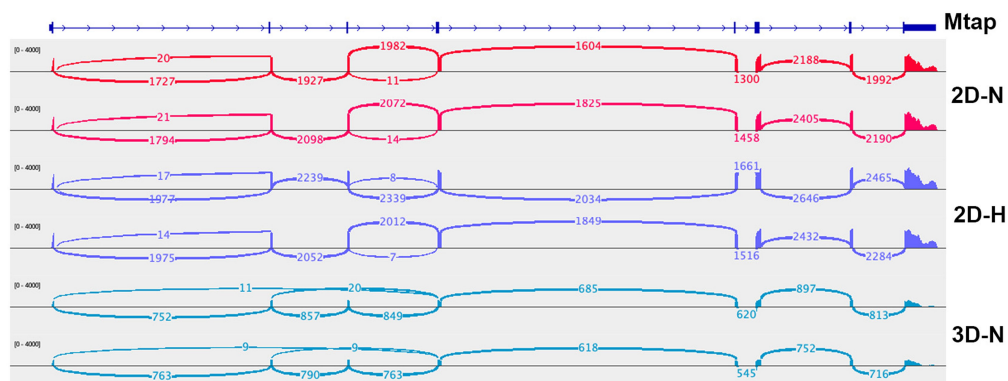


Figure 7- figure supplement 3.

Splicing events of *Mtap* gene under different culture conditions.

Sashimi plot showing exon junction reads of *Mtap* pre-mRNA from RNA-seq data of NEJF10 cells under 2D normoxia, 2D hypoxia and 3D normoxia.

Acknowledgements

We thank the staff of the St. Jude Center for In Vivo Imaging and Therapeutics and Hartwell Center for their dedication and expertise. We thank Dr. Dongli Hu for mycoplasma testing and STR assay. This work was partly supported by American Cancer Society-Research Scholar (130421-RSG-17-071-01-TBG, J.Y.) and National Cancer Institute (1R03CA212802-01A1, 1R01CA229739-01, 1R01CA266600-01A1, J.Y.). This work was supported by the American Lebanese Syrian Associated Charities (ALSAC). The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health. The authors have declared that no conflict of interest exists.

References

1. Kaelin WG (2005) **The concept of synthetic lethality in the context of anticancer therapy** *Nat Rev Cancer* **5**:689–698 <https://doi.org/10.1038/nrc1691>
2. Tsherniak A, Vazquez F, Montgomery PG, Weir BA, Kryukov G, Cowley GS, Gill S, Harrington WF, Pantel S, Krill-Burger JM, et al. (2017) **Defining a Cancer Dependency Map** *Cell* **170**:564–576 <https://doi.org/10.1016/j.cell.2017.06.010>
3. McDonald ER, de Weck A, Schlabach MR, Billy E, Mavrakis KJ, Hoffman GR, Belur D, Castelletti D, Frias E, Gampa K, et al. (2017) **Project DRIVE: A Compendium of Cancer Dependencies and Synthetic Lethal Relationships Uncovered by Large-Scale** *Deep RNAi Screen Cell* **170**:577–592 <https://doi.org/10.1016/j.cell.2017.07.005>
4. Dharia NV, Kugener G, Guenther LM, Malone CF, Durbin AD, Hong AL, Howard TP, Bandopadhyay P, Wechsler CS, Fung I, et al. (2021) **A first-generation pediatric cancer dependency map** *Nat Genet* **53**:529–538 <https://doi.org/10.1038/s41588-021-00819-w>
5. Sun CX, Daniel P, Bradshaw G, Shi H, Loi M, Chew N, Parackal S, Tsui V, Liang Y, Koptyra M, et al. (2023) **Generation and multi-dimensional profiling of a childhood cancer cell line atlas defines new therapeutic opportunities** *Cancer Cell* **41**:660–677 <https://doi.org/10.1016/j.ccell.2023.03.007>
6. Behan FM, Iorio F, Picco G, Goncalves E, Beaver CM, Migliardi G, Santos R, Rao Y, Sassi F, Pinnelli M, et al. (2019) **Prioritization of cancer therapeutic targets using CRISPR-Cas9 screens** *Nature* **568**:511–516 <https://doi.org/10.1038/s41586-019-1103-9>
7. Ungricht R, Guibbal L, Lasbennes MC, Orsini V, Beibel M, Waldt A, Cuttat R, Carbone W, Basler A, Roma G, et al. (2022) **Genome-wide screening in human kidney organoids identifies developmental and disease-related aspects of nephrogenesis** *Cell Stem Cell* **29**:160–175 <https://doi.org/10.1016/j.stem.2021.11.001>
8. Murakami K, Terakado Y, Saito K, Jomen Y, Takeda H, Oshima M, Barker N (2021) **A genome-scale CRISPR screen reveals factors regulating Wnt-dependent renewal of mouse gastric epithelial cells** *Proc Natl Acad Sci U S A* **118** <https://doi.org/10.1073/pnas.2016806118>
9. Ringel T, Frey N, Ringnalda F, Janjuha S, Cherkaoui S, Butz S, Srivatsa S, Pirkl M, Russo G, Villiger L, et al. (2020) **Genome-Scale CRISPR Screening in Human Intestinal Organoids Identifies Drivers of TGF-beta Resistance** *Cell Stem Cell* **26**:431–440 <https://doi.org/10.1016/j.stem.2020.02.007>
10. Takeda H, Kataoka S, Nakayama M, Ali MAE, Oshima H, Yamamoto D, Park JW, Takegami Y, An T, Jenkins NA, et al. (2019) **CRISPR-Cas9-mediated gene knockout in intestinal tumor organoids provides functional validation for colorectal cancer driver genes** *Proc Natl Acad Sci U S A* **116**:15635–15644 <https://doi.org/10.1073/pnas.1904714116>
11. Han K, Pierce SE, Li A, Spees K, Anderson GR, Seoane JA, Lo YH, Dubreuil M, Olivas M, Kamber RA, et al. (2020) **CRISPR screens in cancer spheroids identify 3D growth-specific vulnerabilities** *Nature* **580**:136–141 <https://doi.org/10.1038/s41586-020-2099-x>

12. Thomas LW, Esposito C, Morgan RE, Price S, Young J, Williams SP, Maddalena LA, McDermott U, Ashcroft M (2021) **Genome-wide CRISPR/Cas9 deletion screen defines mitochondrial gene essentiality and identifies routes for tumour cell viability in hypoxia** *Commun Biol* **4**:615 <https://doi.org/10.1038/s42003-021-02098-x>
13. Jain IH, Calvo SE, Markhard AL, Skinner OS, To TL, Ast T, Mootha VK (2020) **Genetic Screen for Cell Fitness in High or Low Oxygen Highlights Mitochondrial and Lipid Metabolism** *Cell* **181**:716–727 <https://doi.org/10.1016/j.cell.2020.03.029>
14. Yang J, Ledaki I, Turley H, Gatter KC, Montero JC, Li JL, Harris AL (2009) **Role of hypoxia-inducible factors in epigenetic regulation via histone demethylases** *Ann N Y Acad Sci* **1177**:185–197 <https://doi.org/10.1111/j.1749-6632.2009.05027.x>
15. Gan ES, Ooi EE (2020) **Oxygen: viral friend or foe?** *Virology* **17**:115 <https://doi.org/10.1186/s12985-020-01374-2>
16. Thannickal VJ (2009) **Oxygen in the evolution of complex life and the price we pay** *Am J Respir Cell Mol Biol* **40**:507–510 <https://doi.org/10.1165/rcmb.2008-0360PS>
17. Lee P, Chandel NS, Simon MC (2020) **Cellular adaptation to hypoxia through hypoxia inducible factors and beyond** *Nat Rev Mol Cell Biol* **21**:268–283 <https://doi.org/10.1038/s41580-020-0227-y>
18. Kaelin WG, Ratcliffe PJ (2008) **Oxygen sensing by metazoans: the central role of the HIF hydroxylase pathway** *Mol Cell* **30**:393–402 <https://doi.org/10.1016/j.molcel.2008.04.009>
19. Chakraborty AA, Laukka T, Myllykoski M, Ringel AE, Booker MA, Tolstorukov MY, Meng YJ, Meier SR, Jennings RB, Creech AL, et al. (2019) **Histone demethylase KDM6A directly senses oxygen to control chromatin and cell fate** *Science* **363**:1217–1222 <https://doi.org/10.1126/science.aaw1026>
20. Liang Y, Ruan W, Jiang Y, Smalling R, Yuan X, Eltzschig HK (2023) **Interplay of hypoxia-inducible factors and oxygen therapy in cardiovascular medicine** *Nat Rev Cardiol* <https://doi.org/10.1038/s41569-023-00886-y>
21. Singleton DC, Macann A, Wilson WR (2021) **Therapeutic targeting of the hypoxic tumour microenvironment** *Nat Rev Clin Oncol* **18**:751–772 <https://doi.org/10.1038/s41571-021-00539-4>
22. Pelullo M, Zema S, Nardoza F, Checquolo S, Screpanti I, Bellavia D (2019) **Wnt, Notch, and TGF-beta Pathways Impinge on Hedgehog Signaling Complexity: An Open Window on Cancer** *Front Genet* **10**:711 <https://doi.org/10.3389/fgene.2019.00711>
23. Egeblad M, Nakasone ES, Werb Z (2010) **Tumors as organs: complex tissues that interface with the entire organism** *Dev Cell* **18**:884–901 <https://doi.org/10.1016/j.devcel.2010.05.012>
24. Harris AL (2002) **Hypoxia-a key regulatory factor in tumour growth** *Nat Rev Cancer* **2**:38–47 <https://doi.org/10.1038/nrc704>
25. Gavish A, Tyler M, Greenwald AC, Hoefflin R, Simkin D, Tschernichovsky R, Galili Darnell N, Somech E, Barbolin C, Antman T, et al. (2023) **Hallmarks of transcriptional intratumour heterogeneity across a thousand tumours** *Nature* **618**:598–606 <https://doi.org/10.1038/s41586-023-06130-4>

26. Fang J, Singh S, Cheng C, Natarajan S, Sheppard H, Abu-Zaid A, Durbin AD, Lee HW, Wu Q, Steele J, et al. (2023) **Genome-wide mapping of cancer dependency genes and genetic modifiers of chemotherapy in high-risk hepatoblastoma** *Nat Commun* **14**:4003 <https://doi.org/10.1038/s41467-023-39717-6>
27. Sanchez-Vega F, Mina M, Armenia J, Chatila WK, Luna A, La KC, Dimitriadou S, Liu DL, Kantheti HS, Saghafein S, et al. (2018) **Oncogenic Signaling Pathways in The Cancer Genome Atlas** *Cell* **173**:321–337 <https://doi.org/10.1016/j.cell.2018.03.035>
28. Knijnenburg TA, Wang L, Zimmermann MT, Chambwe N, Gao GF, Cherniack AD, Fan H, Shen H, Way GP, Greene CS, et al. (2018) **Genomic and Molecular Landscape of DNA Damage Repair Deficiency across The Cancer Genome Atlas** *Cell Rep* **23**:239–254 <https://doi.org/10.1016/j.celrep.2018.03.076>
29. Biamonti G, Caceres JF (2009) **Cellular stress and RNA splicing** *Trends Biochem Sci* **34**:146–153 <https://doi.org/10.1016/j.tibs.2008.11.004>
30. Liu L, Wise DR, Diehl JA, Simon MC (2008) **Hypoxic reactive oxygen species regulate the integrated stress response and cell survival** *J Biol Chem* **283**:31153–31162 <https://doi.org/10.1074/jbc.M805056200>
31. Rios-Lopez DG, Tecalco-Cruz AC, Martinez-Pastor D, Sosa-Garrocho M, Tapia-Urzuza G, Aranda-Lopez Y, Ortega-Dominguez B, Recillas-Targa F, Vazquez-Victorio G, Macias-Silva M (2023) **TGF-beta/SMAD canonical pathway induces the expression of transcriptional cofactor TAZ in liver cancer cells** *Heliyon* **9**:e21519 <https://doi.org/10.1016/j.heliyon.2023.e21519>
32. Li W, Xu H, Xiao T, Cong L, Love MI, Zhang F, Irizarry RA, Liu JS, Brown M, Liu XS (2014) **MAGECK enables robust identification of essential genes from genome-scale CRISPR/Cas9 knockout screens** *Genome Biol* **15**:554 <https://doi.org/10.1186/s13059-014-0554-4>
33. Ashton TM, McKenna WG, Kunz-Schughart LA, Higgins GS (2018) **Oxidative Phosphorylation as an Emerging Target in Cancer Therapy** *Clin Cancer Res* **24**:2482–2490 <https://doi.org/10.1158/1078-0432.CCR-17-3070>
34. Bissell DM, Anderson KE, Bonkovsky HL (2017) **Porphyrin** *N Engl J Med* pp. 862–872 <https://doi.org/10.1056/NEJMra1608634>
35. Fontanellas A, Avila MA (2022) **Hydroxymethylbilane synthase (aka porphobilinogen deaminase): A novel metabolic tumor suppressor gene in hepatocellular carcinoma** *J Hepatol* **77**:912–914 <https://doi.org/10.1016/j.jhep.2022.06.023>
36. Molina L, Zhu J, Trepo E, Bayard Q, Amadeo G, Consortium G, Blanc JF, Calderaro J, Ma X, Zucman-Rossi J, Letouze E (2022) **Bi-allelic hydroxymethylbilane synthase inactivation defines a homogenous clinico-molecular subtype of hepatocellular carcinoma** *J Hepatol* **77**:1038–1046 <https://doi.org/10.1016/j.jhep.2022.05.018>
37. Korkut A, Zaidi S, Kanchi RS, Rao S, Gough NR, Schultz A, Li X, Lorenzi PL, Berger AC, Robertson G, et al. (2018) **A Pan-Cancer Analysis Reveals High-Frequency Genetic Alterations in Mediators of Signaling by the TGF-beta Superfamily** *Cell Syst* **7**:422–437 <https://doi.org/10.1016/j.cels.2018.08.010>
38. Hernandez-Gerez E, Dall'Angelo S, Collinson JM, Fleming IN, Parson SH (2020) **Widespread tissue hypoxia dysregulates cell and metabolic pathways in SMA** *Ann Clin Transl Neurol* **7**:1580–1593 <https://doi.org/10.1002/acn3.51134>

39. Orpinell M, Fournier M, Riss A, Nagy Z, Krebs AR, Frontini M, Tora L (2010) **The ATAC acetyl transferase complex controls mitotic progression by targeting non-histone substrates** *EMBO J* **29**:2381–2394 <https://doi.org/10.1038/emboj.2010.125>
40. Yap TA, Daver N, Mahendra M, Zhang J, Kamiya-Matsuoka C, Meric-Bernstam F, Kantarjian HM, Ravandi F, Collins ME, Francesco MED, et al. (2023) **Complex I inhibitor of oxidative phosphorylation in advanced solid tumors and acute myeloid leukemia: phase I trials** *Nat Med* **29**:115–126 <https://doi.org/10.1038/s41591-022-02103-8>
41. Trigos AS, Pearson RB, Papenfuss AT, Goode DL (2018) **How the evolution of multicellularity set the stage for cancer** *Br J Cancer* **118**:145–152 <https://doi.org/10.1038/bjc.2017.398>
42. Trigos AS, Pearson RB, Papenfuss AT, Goode DL (2017) **Altered interactions between unicellular and multicellular genes drive hallmarks of transformation in a diverse range of solid tumors** *Proc Natl Acad Sci U S A* **114**:6406–6411 <https://doi.org/10.1073/pnas.1617743114>
43. Chen H, Lin F, Xing K, He X (2015) **The reverse evolution from multicellularity to unicellularity during carcinogenesis** *Nat Commun* **6**:6367 <https://doi.org/10.1038/ncomms7367>
44. Rao RC, Dou Y (2015) **Hijacked in cancer: the KMT2 (MLL) family of methyltransferases** *Nat Rev Cancer* **15**:334–346 <https://doi.org/10.1038/nrc3929>
45. Astolfi A, Fiore M, Melchionda F, Indio V, Bertuccio SN, Pession A (2019) **BCOR involvement in cancer** *Epigenomics* **11**:835–855 <https://doi.org/10.2217/epi-2018-0195>
46. Raj N, Wang M, Seoane JA, Zhao RL, Kaiser AM, Moonie NA, Demeter J, Boutelle AM, Kerr CH, Mulligan AS, et al. (2022) **The Mettl3 epitranscriptomic writer amplifies p53 stress responses** *Mol Cell* **82**:2370–2384 <https://doi.org/10.1016/j.molcel.2022.04.010>
47. Wei T, Li J, Zhang J, Zhang Q, Liu X, Chen Q, Wen L, Ma K, Chen W, Zhao J, et al. (2023) **Loss of Mettl3 enhances liver tumorigenesis by inducing hepatocyte dedifferentiation and hyperproliferation** *Cell Rep* **42**:112704 <https://doi.org/10.1016/j.celrep.2023.112704>
48. Hou Y, Zhang X, Yao H, Hou L, Zhang Q, Tao E, Zhu X, Jiang S, Ren Y, Hong X, et al. (2023) **METTL14 modulates glycolysis to inhibit colorectal tumorigenesis in p53-wild-type cells** *EMBO Rep* **24**:e56325 <https://doi.org/10.15252/embr.202256325>
49. Yang Z, Yang S, Cui YH, Wei J, Shah P, Park G, Cui X, He C, He YY (2021) **METTL14 facilitates global genome repair and suppresses skin tumorigenesis** *Proc Natl Acad Sci U S A* **118** <https://doi.org/10.1073/pnas.2025948118>
50. Lombardi O, Li R, Halim S, Choudhry H, Ratcliffe PJ, Mole DR (2022) **Pan-cancer analysis of tissue and single-cell HIF-pathway activation using a conserved gene signature** *Cell Rep* **41**:111652 <https://doi.org/10.1016/j.celrep.2022.111652>
51. Yoda A, Adelmant G, Tamburini J, Chapuy B, Shindoh N, Yoda Y, Weigert O, Kopp N, Wu SC, Kim SS, et al. (2015) **Mutations in G protein beta subunits promote transformation and kinase inhibitor resistance** *Nat Med* **21**:71–75 <https://doi.org/10.1038/nm.3751>

52. Ackerman D, Simon MC (2014) **Hypoxia, lipids, and cancer: surviving the harsh tumor microenvironment** *Trends Cell Biol* **24**:472–478 <https://doi.org/10.1016/j.tcb.2014.06.001>
53. Hall Z, Wilson CH, Burkhart DL, Ashmore T, Evan GI, Griffin JL (2020) **Myc linked to dysregulation of cholesterol transport and storage in nonsmall cell lung cancer** *J Lipid Res* **61**:1390–1399 <https://doi.org/10.1194/jlr.RA120000899>
54. Shachaf CM, Perez OD, Youssef S, Fan AC, Elchuri S, Goldstein MJ, Shirer AE, Sharpe O, Chen J, Mitchell DJ, et al. (2007) **Inhibition of HMGCoA reductase by atorvastatin prevents and reverses MYC-induced lymphomagenesis** *Blood* **110**:2674–2684 <https://doi.org/10.1182/blood-2006-09-048033>
55. Koh CM, Bezzi M, Low DH, Ang WX, Teo SX, Gay FP, Al-Haddawi M, Tan SY, Osato M, Sabo A, et al. (2015) **MYC regulates the core pre-mRNA splicing machinery as an essential step in lymphomagenesis** *Nature* **523**:96–100 <https://doi.org/10.1038/nature14351>
56. Kryukov GV, Wilson FH, Ruth JR, Paulk J, Tsherniak A, Marlow SE, Vazquez F, Weir BA, Fitzgerald ME, Tanaka M, et al. (2016) **MTAP deletion confers enhanced dependency on the PRMT5 arginine methyltransferase in cancer cells** *Science* **351**:1214–1218 <https://doi.org/10.1126/science.aad5214>
57. Appleby TC, Erion MD, Ealick SE (1999) **The structure of human 5'-deoxy-5'-methylthioadenosine phosphorylase at 1.7 Å resolution provides insights into substrate binding and catalysis** *Structure* **7**:629–641 [https://doi.org/10.1016/s0969-2126\(99\)80084-7](https://doi.org/10.1016/s0969-2126(99)80084-7)
58. Li W, Koster J, Xu H, Chen CH, Xiao T, Liu JS, Brown M, Liu XS (2015) **Quality control, modeling, and visualization of CRISPR screens with MAGECK-VISPR** *Genome Biol* **16**:281 <https://doi.org/10.1186/s13059-015-0843-6>
59. Li H, Durbin R (2010) **Fast and accurate long-read alignment with Burrows-Wheeler transform** *Bioinformatics* **26**:589–595 <https://doi.org/10.1093/bioinformatics/btp698>
60. Li H, Handsaker B, Wysoker A, Fennell T, Ruan J, Homer N, Marth G, Abecasis G, Durbin R, 1000 Genome Project Data Processing Subgroup (2009) **The Sequence Alignment/Map format and SAMtools** *Bioinformatics* **25**:2078–2079 <https://doi.org/10.1093/bioinformatics/btp352>
61. Buenrostro JD, Giresi PG, Zaba LC, Chang HY, Greenleaf WJ (2013) **Transposition of native chromatin for fast and sensitive epigenomic profiling of open chromatin, DNA-binding proteins and nucleosome position** *Nat Methods* **10**:1213–1218 <https://doi.org/10.1038/nmeth.2688>
62. Zhang Y, Liu T, Meyer CA, Eeckhoute J, Johnson DS, Bernstein BE, Nusbaum C, Myers RM, Brown M, Li W, Liu XS (2008) **Model-based analysis of ChIP-Seq (MACS)** *Genome Biol* **9**:R137 <https://doi.org/10.1186/gb-2008-9-9-r137>
63. Quinlan AR, Hall IM (2010) **BEDTools: a flexible suite of utilities for comparing genomic features** *Bioinformatics* **26**:841–842 <https://doi.org/10.1093/bioinformatics/btq033>
64. Heinz S, Benner C, Spann N, Bertolino E, Lin YC, Laslo P, Cheng JX, Murre C, Singh H, Glass CK (2010) **Simple combinations of lineage-determining transcription factors prime cis-regulatory elements required for macrophage and B cell identities** *Mol Cell* **38**:576–589 <https://doi.org/10.1016/j.molcel.2010.05.004>

65. Thorvaldsdottir H, Robinson JT, Mesirov JP (2013) **Integrative Genomics Viewer (IGV): highperformance genomics data visualization and exploration** *Brief Bioinform* **14**:178–192 <https://doi.org/10.1093/bib/bbs017>

Author information

Jun Yang

St. Jude Children's Research Hospital Memphis United States
ORCID iD: [0000-0002-0770-9659](https://orcid.org/0000-0002-0770-9659)

For correspondence: Jun.yang2@stjude.org

Jie Fang

St. Jude Children's Research Hospital Memphis United States

Shivendra Singh

St. Jude Children's Research Hospital Memphis United States

Brennan Wells

St. Jude Children's Research Hospital Memphis United States

Qiong Wu

St. Jude Children's Research Hospital Memphis United States
ORCID iD: [0000-0002-6063-0800](https://orcid.org/0000-0002-6063-0800)

Hongjian Jin

St. Jude Children's Research Hospital Memphis United States

Laura Janke

St. Jude Children's Research Hospital Memphis United States

Shibiao Wan

University of Nebraska Medical Center Omaha United States
ORCID iD: [0000-0003-0661-2684](https://orcid.org/0000-0003-0661-2684)

Jacob Steele

St. Jude Children's Research Hospital Memphis United States
ORCID iD: [0000-0001-9924-2226](https://orcid.org/0000-0001-9924-2226)

Jon Connelly

St. Jude Children's Research Hospital Memphis United States
ORCID iD: [0000-0002-5564-1775](https://orcid.org/0000-0002-5564-1775)

Andrew Murphy

St. Jude Children's Research Hospital Memphis United States
ORCID iD: [0000-0001-6747-0355](https://orcid.org/0000-0001-6747-0355)

Ruoning Wang

Abigail Wexner Research Institute at Nationwide Children's Hospital Columbus United States
ORCID iD: [0000-0001-9798-8032](https://orcid.org/0000-0001-9798-8032)

Andrew Davidoff

St. Jude Children's Research Hospital Memphis United States
ORCID iD: [0000-0001-7900-2794](https://orcid.org/0000-0001-7900-2794)

Margaret Ashcroft

University of Cambridge Cambridge United Kingdom
ORCID iD: [0000-0002-0066-3707](https://orcid.org/0000-0002-0066-3707)

Shondra Pruett-Miller

St. Jude Children's Research Hospital Memphis United States
ORCID iD: [0000-0002-3793-585X](https://orcid.org/0000-0002-3793-585X)

Editors

Reviewing Editor

Dean Felsher

Stanford University, Stanford, United States of America

Senior Editor

Wafik El-Deiry

Brown University, Providence, United States of America

Reviewer #1 (Public review):

Summary:

In this report, the authors made use of a murine cell line derived from a MYC-driven liver cancer to investigate the gene expression changes that accompany the switch from normoxic to hypoxia conditions during 2D growth and the switch from 2D monolayer to 3D organoid growth under normoxic conditions. They find a significant (ca. 40-50%) overlap among the genes that are dysregulated in response to hypoxia in 2D cultures and in response to spheroid formation. Unsurprisingly, hypoxia-related genes were among the most prominently deregulated under both sets of conditions. Many other pathways pertaining to metabolism, splicing, mitochondrial electron transport chain structure and function, DNA damage recognition/repair and lipid biosynthesis were also identified.

Comments on the revised manuscript:

In my original review of this manuscript, I raised 11 points that I thought needed to be addressed and/or clarified by the authors. In response, they have provided an adequate answer to only one of these (point 6), which is little more than a more thorough description of how spheroids were generated. The remaining points that I raised, which would have provided more mechanistic insight into their study were addressed by the authors with the following such comments:

- It is not the focus of this study (Points 1 and 4)
- It is worthy of further validation (Point 2)
- We apologize for not being able to validate everything (Point 3)

- This reviewer has raised an interesting question. We are investigating this hypothesis and hopefully we can give a clear answer in the future (Point 5)

- This is an excellent idea that we certainly will do it in our future experiments (Point 7)

As to responses that the authors made to the other two reviewers' comments: Most pertained to cosmetic alterations involving clarification of methods, inclusion of a new figure or rearrangement of old figures. These were generally answered. However, in response to the last point raised by Rev. 3 to compare "sgRNA abundances at the earliest harvesting time with the distribution in the library...to see whether and to what extent selection has already taken place before the three culture conditions were established", the authors responded with the comment: "This is great point. Unfortunately, we did not perform such an analysis."

I understand that it is often impossible to address all points raised by the reviewers. This can be for a variety of reasons and many times the omissions can be overlooked and accepted if the reviewer can be convinced that a good faith attempt has otherwise been made to address the other deficiencies. However, no such effort has been made here and the study remains deficient and largely descriptive as I pointed out in my original review.

<https://doi.org/10.7554/eLife.101299.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

In this report, the authors made use of a murine cell line derived from a MYC-driven liver cancer to investigate the gene expression changes that accompany the switch from normoxic to hypoxia conditions during 2D growth and the switch from 2D monolayer to 3D organoid growth under normoxic conditions. They find a significant (ca. 40-50%) overlap among the genes that are dysregulated in response to hypoxia in 2D cultures and in response to spheroid formation. Unsurprisingly, hypoxia-related genes were among the most prominently deregulated under both sets of conditions. Many other pathways pertaining to metabolism, splicing, mitochondrial electron transport chain structure and function, DNA damage recognition/repair, and lipid biosynthesis were also identified.

We thank this reviewer for his/her time and efforts, and the insightful comments.

Major comments:

(1) Lines 239-240: The authors state that genes involved in DNA repair were identified as being necessary to maintain survival of both 2D and 3D cultures (Figure S6A). Hypoxia is a strong inducer of ROS. Thus, the ROS-specific DNA damage/recognition/repair pathways might be particularly important. The authors should look more carefully at the various subgroups of the many genes that are involved in DNA repair. They should also obtain at least a qualitative assessment of ROS and ROS-mediated DNA damage by staining for total and mitochondrial-specific ROS using dyes such as CM-H2-DCFDA and MitoSox. Actual direct oxidative damage could be assessed by immunostaining for 8-oxo-dG and related to the sub-types of DNA damage-repair genes that are induced. The

centrality of DNA damage genes also raises the question as to whether the previously noted prominence of the TP53 pathway (see point 5 below) might represent a response to ROS-induced DNA damage.

We thank this reviewer for the insightful comments, and agreed that ROS induced by hypoxia could play a role in modulating DNA repair and consequently cellular essentiality. Although pathway enrichment in Figure S6A (now as Figure 2-figure supplement 4A) showed that DNA repair pathway was essential to cell survival in hypoxia and 3D cultures, the genes associated with this pathway (Ddb1;Brf2;Gtf3c5;Guk1;Taf6) are not typical DNA repair genes. They are more likely involved in gene transcription. However, it will be interesting to see if they are specifically involved in DNA damage in response to ROS, which is out of focus of this study.

(2) Because most of the pathway differences that distinguish the various cell states from one another are described only in terms of their transcriptome variations, it is not always possible to understand what the functional consequences of these changes actually are. For example, the authors report that hypoxia alters the expression of genes involved in PDH regulation but this is quite vague and not backed up with any functional or empirical analyses. PDH activity is complex and regulated primarily via phosphorylation/dephosphorylation (usually mediated by PDK1 and PDP2, respectively), which in turn are regulated by prevailing levels of ATP and ADP. Functionally, one might expect that hypoxia would lead to the down-regulation of PDH activity (i.e. increased PDH-pSer392) as respiration changes from oxidative to non-oxidative. This would not be appreciated simply by looking at PDH transcript levels. This notion could be tested by looking at total and phospho-PDH by western blotting and/or by measuring actual PDH activity as it converts pyruvate to AcCoA.

We agreed with this reviewer that PDH activity regulation could be affected by multi-factors, and it is worthy of further validation by other approaches.

(3) Line 439: Related to the above point: the authors state: "It is likely that blockade of acetyl-CoA production by PDH knockout may force cells to use alternative energy sources under hypoxic and 3D conditions, averting the Warburg effect and promoting cell survival under limited oxygen and nutrient availability in 3D spheroids." This could easily be tested by determining whether exogenous fatty acids are more readily oxidized by hypoxic 2D cultures or spheroids than occurs in normoxic 2D cultures.

We thank for this suggestion. We apologized for not being able to validate everything.

(4) Line 472: "Hypoxia induces high expression of Acaca and Fasn in NEJF10 cells indicating that hypoxia promotes saturated fatty acid synthesis...The beneficial effect of Fasn and Acaca KO to NEJF10 under hypoxia is probably due to reduction of saturated fatty acid synthesis, and this hypothesis needs to be tested in the future.". As with the preceding comment, this supposition could readily be supported directly by, for example, performing westerns blots for these enzymes and by showing that incubation of hypoxic 2D cells or spheroids converted more AcCoA into lipid.

We thank for this suggestion. However, functional validation for the Fasn and Acaca KO is out of focus in this study.

(5) In Supplementary Figure 2B&C, the central hub of the 2D normoxic cultures is Myc (as it should well be) whereas, in the normoxic 3D, the central hub is TP53 and Myc is not even present. The authors should comment on this. One would assume that Myc levels should still be quite high given that Myc is driven by an exogenous promoter. Does the

centrality of TP53 indicate that the cells within the spheroids are growth-arrested, being subjected to DNA damage and/or undergoing apoptosis?

The predicted transcription factor activity analysis was based on the differential ATAC-seq peaks among different culture through pairwise comparisons. If TP53 and MYC were not present under that condition, it did not mean their activity was absent.

“...the centrality of TP53 indicate that the cells within the spheroids are growth-arrested, being subjected to DNA damage and/or undergoing apoptosis?” This reviewer has raised an interesting question. We are investigating this hypothesis and hopefully we can give a clear answer in the future.

(6) In the Materials and Methods section (lines 711-720), the description of how spheroid formation was achieved is unclear. Why were the cells first plated into non-adherent 96 well plates and then into nonadherent T75 flasks? Did the authors actually utilize and expand the cells from 144 T75 flasks and did the cells continue to proliferate after forming spheroids? Many cancer cell types will initially form monolayers when plated onto non-adherent surfaces such as plastic Petri dishes and will form spheroid-like structures only after several days. Other cells will only aggregate on the "non-adherent" surface and form spheroid-like structures but will not actually detach from the plate's surface. Have the authors actually documented the formation of true, non-adherent spheroids at 2 days and did they retain uniform size and shape throughout the collection period? The single photo in Supplementary Figure 1 does not explain when this was taken. The authors include a schematic in Figure 2A of the various conditions that were studied. A similar cartoon should be included to better explain precisely how the spheroids were generated and clarify the rationale for 96 well plating. Overall, a clearer and more concise description of how spheroids were actually generated and their appearance at different stages of formation needs to be provided.

The cells were initially plated in non-adherent 96-well plates to facilitate the formation of spheroids in a controlled and uniform manner. As correctly mentioned by the reviewer, during the initial stages, cells cultured on non-adherent surfaces often form aggregates or clumps, and it takes a few days for them to develop into solid spheroids.

In our study, we aimed to achieve 3D spheroid formation immediately following the transduction process to allow for screening under both 2D and 3D conditions. Plating the cells into 96-well plates enabled us to monitor and control the formation of spheroids in smaller volumes before scaling up the culture in non-adherent T75 flasks for subsequent experimental steps. This setup allows us to maintain gene editing processes under both 2D and 3D conditions.

Regarding the proliferation and uniformity of spheroids:

- Yes, the spheroids continued to proliferate after their formation.
- True, non-adherent spheroids were documented as early as the next day. This was visually confirmed under microscopy, and size uniformity was maintained throughout the collection period by following optimized culture protocols.

We also agreed with the reviewer's suggestion to include a cartoon schematic similar to Figure 2A, illustrating the spheroid generation process and clarifying the rationale for using 96-well plates. We have included such a cartoon and spheroid growth curve monitored by Incucyte as Figure 2-figure supplement 2.

(7) The authors maintained 2D cultures in either normoxic or hypoxic (1% O₂) states during the course of their experiments. On the other hand, 3D cultures were maintained

under normoxic conditions, with the assumption that the interiors of the spheroids resemble the hypoxic interiors of tumors. However, the actual documentation of intra-spheroid hypoxia is never presented. It would be a good idea for the authors to compare the degree of hypoxia achieved by 2D (1% O₂) and 3D cultures by staining with a hypoxia-detecting dye such as Image-iT Green. Comparing the fluorescence intensities in 2D cultures at various O₂ concentrations might even allow for the construction of a "standard curve" that could serve to approximate the actual internal O₂ concentration of spheroids. This would allow the authors to correlate the relative levels of hypoxia between 2D and 3D cultures.

This is an excellent idea that we certainly will do it in our future experiments.

(8) Related to the previous 2 points, the authors performed RNAseq on spheroids only 48 hours after initiating 3D growth. I am concerned that this might not have been a sufficiently long enough time for the cells to respond fully to their hypoxic state, especially given my concerns in Point 6. Might the results have been even more robust had the authors waited longer to perform RNA seq? Why was this short time used?

We agreed with this reviewer. We were unsure if 48hours was an ideal timepoint. It might be necessary to perform a longitudinal experiment to harvest samples under different timepoints in the future experiments.

(9) What happens to the gene expression pattern if spheroids are re-plated into standard tissue culture plates after having been maintained as spheroids? Do they resume 2D growth and does the gene expression pattern change back?

This is a great question and we have never thought about what the gene expression pattern would be if spheroids are re-plated in 2D. This could be a challenging experiment because the gene expression and epigenetic changes are timing related. However, the cells do grow well after re-plated in 2D.

(10) Overall, the paper is quite descriptive in that it lists many gene sets that are altered in response to hypoxia and the formation of spheroids without really delving into the actual functional implications and/or prioritizing the sets. Some of these genes are shown by CRISPR screening to be essential for maintaining viability although in very few cases are these findings ever translated into functional studies (for example, see points 14 above). The list of genes and gene pathways could benefit from a better explanation and prioritization of which gene sets the authors believe to be most important for survival in response to hypoxia and for spheroid formation.

This was a genome-wide study that integrated RNA-seq, ATAC-seq and CRISPR KO, providing resource to understand the oncogenic pathways in different culture conditions. We believe we have clearly articulated the important genes/pathways in our abstract.

(11) The authors used a single MYC-driven tumor cell line for their studies. However, in their original paper (Fang, et al. Nat Commun 2023, 14: 4003.) numerous independent cell lines were described. It would help to know whether RNAseq studies performed on several other similar cell lines gave similar results in terms of up & down-regulated transcripts (i.e. representative of the other cell lines are NEJF10 cells).

We have not generated RNA-seq data for these cell lines cultured in different conditions.

Reviewer #2 (Public review):

Summary:

The manuscript by Fang et al., provides a tour-de-force study uncovering cancer cell's varied dependencies on several gene programs for their survival under different biological contexts. The authors addressed genomic differences in 2D vs 3D cultures and how hypoxia affects gene expression. They used a Myc-driven murine liver cancer model grown in 2D monolayer culture in normoxia and hypoxia as well as cells grown as 3D spheroids and performed CRISPR-based genome-wide KO screen to identify genes that play important roles in cell fitness. Some context-specific gene effects were further validated by in-vitro and in-vivo gene KO experiments.

Strengths:

The key findings in this manuscript are:

- (1) Close to 50% of differentially expressed genes were common between 2D Hypoxia and 3D spheroids conditions but they had differences in chromatin accessibility.*
- (2) VHL-HIF1a pathway had differential cell fitness outcomes under 2D normoxia vs 2D hypoxia and 3D spheroids.*
- (3) Individual components of the mitochondrial respiratory chain complex had contrasting effects on cell fitness under hypoxia.*
- (4) Knockout of organogenesis or developmental pathway genes led to better cell growth specifically in the context of 3D spheroids and knockout of epigenetic modifiers had varied effects between 2D and 3D conditions.*
- (5) Another key program that leads to cells fitness outcomes in normoxia vs hypoxia is the lipid and fatty acid metabolism.*
- (6) Prmt5 is a key essential gene under all growth conditions, but in the context of 3D spheroids even partial loss of Prmt5 has a synthetic lethal effect with Mtap deletion and Mtap is epigenetically silenced specifically in the 3D spheroids.*

We appreciate this reviewer for acknowledging the strengths of our study.

Issues to address:

- (1) The authors should clarify the link between the findings of the enrichment of TGFb-SMAD signaling REACTOME pathway to the findings that knocking out TGFb-SMAD pathway leads to better cell fitness outcomes for cells in the 3D growth conditions.*

We have clarified this link in abstract by saying “Notably, multicellular organogenesis signaling pathways including TGFb-SMAD, which is upregulated in 3D culture, specifically constrict the uncontrolled cell proliferation in 3D while inactivation of epigenetic modifiers (*Bcor*, *Kmt2d*, *Mettl3* and *Mettl14*) has opposite outcomes in 2D vs. 3D:

- (2) Supplementary Figure 4C has been cited in the text but doesn't exist in the supplementary figures section.*

Sorry for this typo. It should be 5C which is Figure 2-figure supplement 3C in the new version of MS. We have corrected it now.

(3) A small figure explaining this ABC-Myc driven liver cancer model in Supplementary Figure 1 would be helpful to provide context.

We appreciate this suggestion. We have added a cartoon as Figure 1-figure supplement 1A to indicate the procedure for generation of this model.

(4) The method for spheroids formation is not found in the method section.

We described the method in our previous publication (Nature Communications 2023 Jul 6;14(1):4003.). However, we have added the information in method now, and the procedure is very simple (line 623-624). We found the murine liver cancer cell lines can readily form spheroids when they are cultured in low-attachment dish with standard DMEM complete media.

(5) In Supplementary Figure 1b, the comparisons should be stated the opposite way - 3D vs 2D normoxia and 2D-Hypoxia vs 2D-Normoxia.

We have made correction in the Figure legend of Figure S1B which is Figure 1B now in the new version of MS.

(6) There are typos in the legend for Supplementary Figure 10.

We have checked the typos.

(7) Consider putting Supplementary Figure 1b into the main Figure 1.

We have moved both Supplementary Figure 1a and 1b into main Figure 1 as Figure 1A and 1B. Hopefully, this will help the readers to catch the information easily.

(8) Please explain only one timepoint (endpoint) for 3D spheroids was performed for the CRISPR KO screen experiment, while several timepoints were done for 2D conditions? Was this for technical convenience?

As this reviewer speculated, indeed this was for technical convenience. We found that it was technically challenging to split the spheroids for CRISPR screening.

(9) In line 372, it is indicated that Bcor KO (Fig 5e) had growth advantage - this was observed in only one of the gRNA -- same with Kmt2d KO in the same figure where there was an opposite effect. Please justify the use of only one gRNA.

We actually used 4 gRNAs for each gene. In the heatmap, although one of the gRNA for each gene showed some levels of enrichment under hypoxic 2D condition, they were all highly enriched in 3D.

(10) Why was CRISPR based KO strategy not used for the PRMT5 gene but rather than the use of shRNA.? Note that one of the shRNA for PRMT5 had almost no KO (PRMT5-shRNA2 Figure 7B) but still showed phenotype (Figure 7D) - please explain.

We used shRNA as second approach for cross-validation. We agreed that the knockdown efficiency of shRNA2 was not as good as the others, with only about 40% knockdown efficiency.

(11) In Figure 7D, which samples (which shRNA group) were being compared to do the t-test?

The comparisons were for shCtrl and each of the shPRMT5. We have clarified this in figure legend.

(12) In line 240, it is stated that oxphos gene set is essential for NEJF10 cell survival in both normoxia and hypoxia conditions. But shouldn't oxphos be non-essential in hypoxia as cells move away from oxphos and become glycolytic?

This is a great question. While indeed hypoxia may promote the switch from oxphos to glycolysis, several studies showed that the low oxygen concentrations in hypoxic regions of tumors may not be limiting for oxphos, and ATP is generated by oxphos in tumors even at very low oxygen tensions (please see review Clin Cancer Res (2018) 24 (11): 2482–2490.). We therefore speculated that NEJF10 cells were still dependent on oxphos for ATP production under hypoxia. However, this needs further investigation. We have added this discussion in our manuscript (line 250-254).

(13) In line 485 it is mentioned that Pmvk and Mvd genes which are involved in cholesterol synthesis when knocked out had a positive effect on cell growth in 3D conditions and since cholesterol synthesis is essential for cell growth how does this not matter much in the context of 3D - please explain.

We thank this reviewer for this note. It seemed that only two gRNA for each were upregulated in 3D and it could be due to technical issue or clonal selection. We have deleted this sentence in our new version of MS.

Reviewer #3 (Public review):

Summary:

In this study, Fang et al. systematically investigate the effects of culture conditions on gene expression, genome architecture, and gene dependency. To do this, they cultivate the murine HCC line NEJF10 under standard culture conditions (2D), then under similar conditions but under hypoxia (1% oxygen, 2D hypoxia) and under normoxia as spheroids (3D). NEJF10 was isolated from a murine HCC model that relies exclusively on MYC as a driver oncogene. In principle, (1) RNA-seq, (2) ATAC-seq and (3) genetic screens were then performed in this isogenic system and the results were systematically compared in the three cultivation methods. In particular, genome-wide screens with the CRISPR library Brie were performed very carefully. For example, in the 2D conditions, many different time points were harvested to control the selection process kinetically. The authors note differential dependencies for metabolic processes (not surprisingly, hypoxia signaling is affected) such as the regulation and activity of mitochondria, but also organogenesis signaling and epigenetic regulation.

Strengths:

The topic is interesting and relevant and the experimental set-up is carefully chosen and meaningful. The paper is well written. While the study does not reveal any major surprises, the results represent an important resource for the scientific community.

We thank this reviewer for his/her positive comments.

Weaknesses:

However, this presupposes that the statistical analysis and processing are carried out very carefully, and this is where my main suggestions for revision begin. Firstly, I cannot find any information on the number of replicates in RNA- and ATAC-seq. This should be clearly stated in the results section and figure legends and cut-offs, statistical procedures, p-values, etc. should be mentioned as well. In principle, all NGS experiments (here ATAC- and RNA-seq) should be performed in replicates (at least duplicates, better triplicates) or the results should be validated by RT-PCR in independent biological triplicates. Secondly, the quantification of the analyses shown in the figures and especially in the legends is not sufficiently careful. Units are often not mentioned. Example Figure 4a: The legend says: 'gRNA reads' but how can the read count be -1? I would guess these are FC, log2FC, or Z-values. All figure legends need careful revision.

Based upon the reviewer's suggestions, we have added details about the replicates in figure legend. For gRNA read heatmap, the scale bar indicates the Z score. We have added the information in figure legends.

Furthermore, I would find a comparison of the sgRNA abundances at the earliest harvesting time with the distribution in the library interesting, to see whether and to what extent selection has already taken place before the three culture conditions were established (minor point).

This is great point. Unfortunately, we did not perform such an analysis.

Recommendations for the authors:

Reviewing Editor:

There are three general issues:

First, there is a lack of detail regarding much of the analysis. In some cases, this makes it difficult to assess the value of the data, albeit, there is generally a consensus the information is really interesting.

Second, the findings - although provocative - lack mechanistic details and are focused more on descriptive findings. Hence, the manuscript would be improved by some effort at evaluating identified programs and providing some suggestions of mechanisms.

Third, the authors need to put much more effort into the clarity and tightness of the presentation.

We have made clarification in response to the reviewer's comments.

Reviewer #1 (Recommendations for the authors):

Figure S1C. the labeling of the lower x-axis is inverted.

Due to space limitation, we changed the figure orientation in our old version of MS. We have tilted the figure back in the new version, which is Figure 1-figure supplement 1B now.

<https://doi.org/10.7554/eLife.101299.2.sa0>